Who Contributes What in Human-AI Research? A Qualitative Case Study of a Project That Analyses Itself

Introduction

Generative AI now takes part in every stage of scholarship, from the formulation of questions to the writing of manuscripts. Information systems (IS) researchers have begun to chart how the field should conduct its own work with it (Susarla et al., 2023). The call for this workshop asks every submission for a human-AI contribution account that covers nine research functions (ICIS 2026 PDW on Autonomous Scholarship, 2026). Current disclosure practice answers with a list of tasks, which shows activity and hides the kind of thinking behind it. A draft by the AI may rest on a scope that the human fixed, and an approval by the human may follow a framing that the AI proposed.

The hidden layer matters for IS theory as well as for disclosure. IS research describes the division of labour between humans and AI mainly through delegation. Baird and Maruping (2021) offer delegation to and from agentic artifacts as a lens, Fügener et al. (2022) show that humans delegate knowledge work to AI poorly, and Seeber et al. (2020) treat machines as teammates. These accounts describe what is handed over and with what result. They leave open what each party thinks, so two widely held worries rest on assertion. One says that humans become unnecessary once AI can execute a research workflow. The other says that AI is an execution tool with no ideas of its own.

This paper asks one plain question about this new kind of work and studies it where the full trace is available, in the project that produced the paper. The question is how humans and AI share the thinking when they do research together. Three sub-questions refine it.

- **RQ1 Existence.** What kinds of thinking does each party contribute, and do some outputs draw on both?
- **RQ2 Passing.** How does the thinking pass between the two as the work proceeds?
- **RQ3 Explanation.** What explains this division and this sequence?

A human orchestrator directed AI agents in the project, and every session was recorded as it happened. The sessions are therefore both the data and the history of the manuscript's own production. We build a typology of observable cognitive contributions and code 265 units with two LLM coders from different model families. Human, AI and mixed contributions all exist. The human holds commitment and steers the process, the AI supplies knowledge, candidate proposals and verification, and seven outputs carry explicit contributions from both. The sequence follows a cycle of delegation, generation and commitment, which we state as three propositions.

The paper makes three contributions to IS research. It offers a typology of observable cognitive contributions that treats knowledge as something an AI can supply. It offers a delegation and commitment account that explains why contributions divide and pass as they do, stated as propositions that other projects can test. It also provides a trace-based coding standard that makes the account inspectable.

Theoretical Background

Human-AI Work in Information Systems Research

IS research frames the division of labour between humans and AI through delegation, configuration and performance. Baird and Maruping (2021) argue that agentic artifacts can take responsibility for tasks with ambiguous requirements and propose delegation to and from them as the organising lens. Fügener et al. (2022) show that humans misjudge their own metaknowledge and delegate knowledge work to AI poorly. Grønsund and Aanestad (2020) trace how humans and an algorithm were configured and reconfigured in one organisation. Seeber et al. (2020) set out a research agenda for machines as teammates, and Berente et al. (2021) name autonomy as a facet that managers of AI must decide. Automation and augmentation are

intertwined (Raisch & Krakowski, 2021). A meta-analysis finds that human-AI combinations on average perform worse than the better party alone, with losses in decision tasks and gains in content creation (Vaccaro et al., 2024).

Trace-based studies of what each party does are fewer. Handa et al. (2025) label conversations as automative or augmentative, and CoAuthor records keystroke-level co-writing in which writers produced 72.6% of the final text (Lee et al., 2022). Survey evidence suggests that effort shifts toward verification, integration and stewardship when workers use generative AI (Lee et al., 2025). In science, AI Co-Scientist assigns generation, critique and refinement to agents while scientists set goals and review results (Gottweis et al., 2026), and Lu et al. (2024) automate a whole machine learning research pipeline. Costa et al. (2025) describe a qualitative workflow in which the AI acts as analytical partner and the human as reflexive agent. These works describe contributions by task, outcome or role. We found no study that types the cognitive kind of each party's contribution across a whole project or explains why the work divides as it does.

Coding schemes for related objects exist. Cognitive presence codes four phases of inquiry at message level among human participants (Garrison et al., 2001). Initiative coding records who holds control and leaves content aside (Walker & Whittaker, 1990). The typology below borrows the indicator logic of the first and the observable stance of the second.

A Typology of Cognitive Contributions

The typology rests on cognition research. Cognitive taxonomies separate processes from knowledge types (Krathwohl, 2002), and knowledge divides into situational, conceptual, procedural and strategic kinds (de Jong & Ferguson-Hessler, 1996). Decision research describes choice as intelligence, design and choice (Simon, 1960), and Mintzberg et al. (1976) found evaluation and choice separable in only 18 of 83 instances. Design research treats externalised structure as part of thinking (Gero & Kannengiesser, 2004), and epistemic actions reduce the burden of mental computation (Kirsh & Maglio, 1994). Metacognition spans knowledge, experiences, goals and strategies (Flavell, 1979), and generative AI adds the demands of prompting and output evaluation (Tankelevitch et al., 2024). People hand cognitive demands to external resources according to a metacognitive evaluation of those demands (Risko & Gilbert, 2016).

Two properties of these sources shape the typology. First, they place knowledge in the thinker, so that evaluation, choice and design operate on what the thinker already holds. When an AI agent supplies knowledge that a human expert did not hold, knowledge becomes something that is contributed, and we treat Knowledge as a type. Second, cognition is inferred from what is observable. Garrison et al. (2001) derived manifest indicators for each latent phase of inquiry, and we classify what a move does in the shared record.

A cognitive contribution is an externalised move by an actor that adds to or changes the shared research state and can be identified from the trace. Figure 1, in the next section, shows the types and the pipeline in which they are used. Knowledge introduces a fact, concept, method or situational context. Proposal offers a candidate, such as a frame, structure, text or revision, that is not yet committed to. Decision commits the project by setting a goal, scope or constraint, or by selecting, rejecting or stopping an option. Metacognition checks or steers how the work is done and how reliable it is. Execution carries out a specified step without new content. Evaluation and choice form one type, since evaluation usually carries preference.

One rule separates object from process. If removing a move would change the product, the move is Knowledge, Proposal or Decision. If it would change how the product was made or checked, the move is Metacognition. Proposals offer and decisions commit, so an AI draft is a Proposal until a human adopts it, and the adoption is a separate Decision. The codebook defines subtypes and boundary rules for each type. Knowledge, for instance, divides into fact or source, concept or explanation, method or procedure, and situated context.

Research Design

Case and Data

The case is the project that produced this manuscript, studied with the case research strategy established for IS (Benbasat et al., 1987). One human orchestrator directed a primary AI agent (claude-sonnet-5.5),

which dispatched sub-agents on the same model to survey and read literature. Every session was exported with message-level provenance, model, agent name and parent session. The corpus comprises 13 sessions, namely the ideation session, the main working session, ten sub-agent sessions and a second primary-agent session that redrew the figures. These yield 265 units, of which 38 are human messages, 177 are primary-agent responses, 10 are primary-agent prompts to sub-agents and 40 are sub-agent responses. The corpus ends at the human message that asked for the recoding reported here (S01-U186). Empty messages, continuation prompts and context summaries that the agent framework inserted were removed (27 units). The designer of the typology also produced most of the units that the typology classifies, so the case is reflexive.

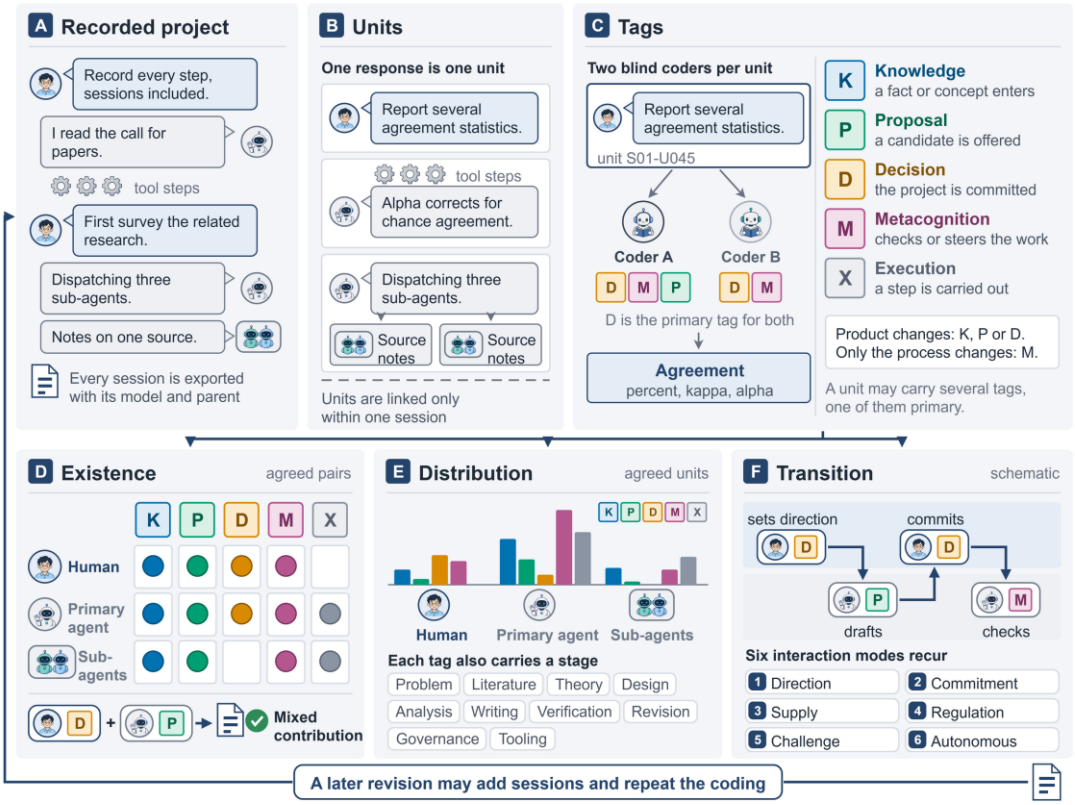


Figure 1. From recorded sessions (A) through units (B) and tags (C) to the analyses of existence (D), distribution (E) and transitions (F), from which the account of delegation and commitment is built. Bubble texts abridge project messages.

Units and Coding

A unit is one model response. A human unit is one human message. An AI unit is an assistant message that contains visible text or dispatches a sub-agent, together with the tool-only steps that precede it. Tool-only steps (999 of 1,225 assistant messages) serve as evidence for the unit that follows. A unit may carry several tags, each with an actor and a type, and at most one primary tag per actor. Each tag carries a stage label drawn from the workshop's research functions plus governance and tooling, and an evidence quotation. The codebook gives each code a definition, rules for use and non-use, and examples (MacQueen et al., 1998).

Two independent coders, GPT-6.1-sol and Gemini-4-argon, coded every unit with the frozen codebook (version 1.1) as their instruction, as in the LLM annotation studies of Gilardi et al. (2023) and Xiao et al. (2023). Each ran in an isolated session with the codebook and the units, in chunks of up to 20. Each received the unit text, truncated at 2,500 characters, and an actor hint with model names masked. The primary agent designed the codebook and coordinated the coding, and the analysis uses the codes of the two coders alone. The human approved each codebook version and ruled on boundary cases. The coding standard is a method artifact in the design-science sense (Hevner et al., 2004).

The coding was run twice. Run 1 coded the first 146 units. When the framing of the manuscript changed, run 2 recoded these units together with 119 later ones, as the project rules require at each larger revision. We report run 2 and state where run 1 differs.

Analysis

We turn each question into observable terms. For RQ1, an actor-type pair counts as present when both coders assign it to at least one unit. A contribution is mixed when both actors add explicit expression to the same output that adds, changes or reasons about content, and bare approvals do not count. For RQ2, we order the cognitive units within each session, write each transition between consecutive units as actor and primary type, and group recurring transitions by content into interaction modes.

For RQ3, we build theory from the case evidence in the manner of case research (Eisenhardt, 1989). We divide each session into delegation episodes, each beginning with a human message and holding the AI responses up to the next human message. We read the episodes against the codes to see what opens and closes a run of AI work, which party commits, and how knowledge changes a construct. The propositions generalise analytically (Lee & Baskerville, 2003). We report percent agreement, Cohen's kappa and Krippendorff's alpha with bootstrap intervals for each actor-type presence variable (Artstein & Poesio, 2008, Hayes & Krippendorff, 2007). We also compare the two runs on the 146 units that both coded.

Findings

What Each Party Contributes (RQ1)

Human contributions occur in all four types (Table 1). Decision appears in 28 of 38 human units, Metacognition in 25, Knowledge in 12 and Proposal in 3. The human chose the topic, restricted the data to the project's own sessions, fixed the unit of analysis as one model response and set the research questions. Human Metacognition regulated the process, for instance by requiring every session to be recorded (S01-U001) and originals to be read (S01-U020). Human Knowledge was mostly situated context that only the human held, such as an image service that the AI had reported as unavailable (S13-U006). Human Proposals were rare and shaped core elements of the design (S01-U014, S01-U032, S01-U070).

Actor (units)	Knowledge	Proposal	Decision	Metacognition	Execution
Human (38)	12 (32%) [12/12]	3 (8%) [3/4]	28 (74%) [32/29]	25 (66%) [26/28]	0 (0%) [0/0]
Primary agent (187)	40 (21%) [99/42]	22 (12%) [23/38]	4 (2%) [23/8]	106 (57%) [113/111]	68 (36%) [68/118]
Sub-agents (40)	10 (25%) [14/10]	1 (3%) [1/1]	0 (0%) [0/0]	12 (30%) [17/12]	19 (48%) [19/32]

Table 1. Units with each type present in both coders' tags, with the count of each coder in brackets (GPT/Gemini).

AI contributions occur in Knowledge, Proposal, Metacognition and Decision. The primary agent supplied Knowledge in 40 of its 187 units, for example an explanation of agreement coefficients (S01-U044). The sub-agents supplied it in 10 of 40 units, almost all of it literature. The primary agent offered Proposals in 22 units, among them candidate taxonomies (S01-U016), research questions (S01-U069) and the figure drafts (S01-U149). Its Metacognition (106 units) consisted of verification results, detected errors and statements about its own limits. Both coders agree on four AI Decision units. Three are task assignments to sub-agents. The fourth is the primary agent's decision not to repeat the coding after a revision (S01-U185), which the human overturned in the next message (S01-U186). The human therefore remained necessary, and the AI did more than execute.

Mixed contributions exist in the project. We identified seven outputs to which both actors contributed with explicit expression (Table 2), and they show four collaboration modes. In the first, the AI supplies knowledge and the human builds a proposal on it. On the AI's literature summary, the human built the proposal to treat knowledge as a type (S01-U032). In the second, the human fixes a decision and the AI turns it into an operational proposal grounded in data, as when the unit of analysis was defined. In the third, the human reformulates an AI proposal. Three research questions replaced the AI's candidate set (S01-

U070), and the figures and the framing of the paper followed the same mode. In the fourth, the human regulates how the AI supplies knowledge, for instance by requiring a survey first and the reading of originals. Two short human approvals, S01-U078 and S01-U080, left the proposed outputs as they were.

Output	AI contribution	Human contribution	Mode
Four-type typology	Literature synthesis (S01-U031), framework draft (S01-U034)	Knowledge as a type, Proposal replaces design (S01-U032)	AI Knowledge, human Proposal
Unit of analysis	Draft rule (S01-U039), operational rule (S01-U042)	One model response as the unit (S01-U040)	Human Decision, AI Proposal
Research questions	Candidate questions (S01-U069, S01-U072)	Three questions with reasons (S01-U070)	Human reformulation
Mixed criterion	Draft criterion (S01-U072)	Same-output rule, no bare approvals (S01-U073)	Human reformulation
Literature base	Surveys and notes (S01-U019, S01-U031)	Survey first, originals, skip closed sources (S01-U017, S01-U020, S01-U032)	Human Metacognition, AI Knowledge
Figures 1 and 2	Whole-figure drafts, rebuilds (S01-U149, S01-U167)	Concrete examples, side tails (S01-U164, S13-U025, S13-U032)	Human reformulation
Framing of the paper	Comparison with IS papers (S01-U174)	Co-author's comment, choice of option A (S01-U168, S01-U175)	Human reformulation

Table 2. Outputs with contributions from both actors and their collaboration mode.

How Contributions Pass Between Parties (RQ2)

Figure 2 groups the transitions between consecutive cognitive units within sessions. The two coders found 165 and 146 transitions, and six interaction modes recur. Direction and drafting (human Decision, then AI Proposal) and proposal and commitment (AI Proposal, then human Decision) form a loop that accounts for 18 and 21 transitions. The human sets or reframes a direction, the AI drafts, and the human commits or redirects. Supply and reframing (AI Knowledge, then human Decision) is rare, with 3 and 4 transitions. Regulation and response (human Decision or Metacognition, then AI Metacognition) accounts for 17 and 17 transitions, and challenge (AI Proposal, then human Metacognition) for 3 and 4. Autonomous work (AI Metacognition continuing or leading to AI Knowledge) is the most frequent mode with 85 and 62 transitions. A report followed by a human Decision occurs 7 and 6 times and lies outside the six modes.

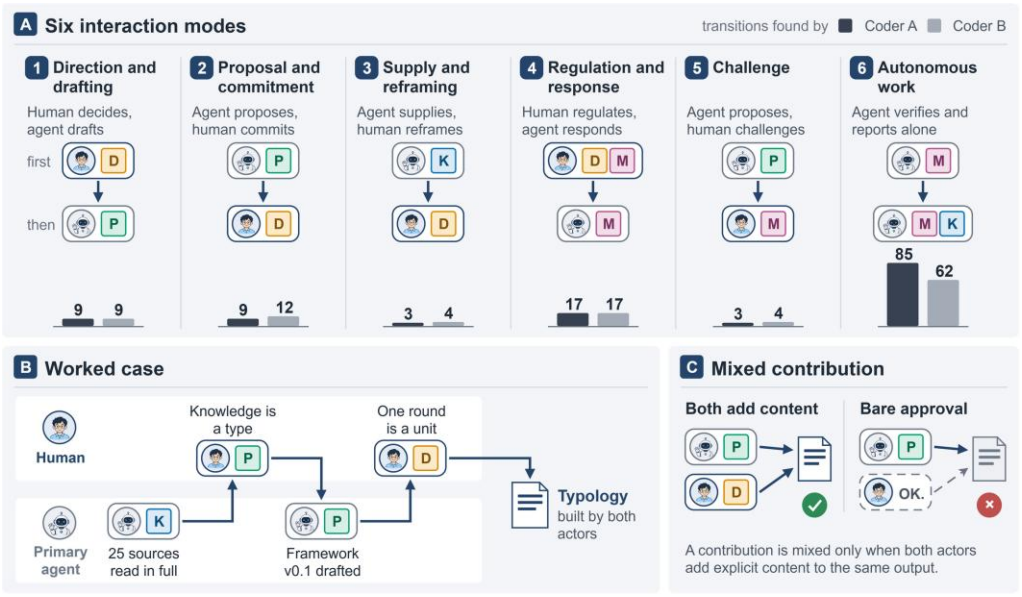


Figure 2. (A) Six interaction modes with the transition counts of Coder A (GPT) and Coder B (Gemini). (B) A worked case with the types both coders tagged. (C) Mixed contribution and bare approval. Bubble texts abridge project messages.

Five of the seven outputs in Table 2 show an AI Proposal, a human Decision and a further AI Proposal in succession. The typology followed a related path in which AI-supplied knowledge preceded a human reframing, an AI draft and a human ruling (S01-U031 to S01-U035). A single pass of proposal and commitment, as in the short approvals of the last rounds, left an output that one actor shaped. The human reaches the AI mainly through direction and regulation, and the AI reaches the human mainly through proposals.

A Delegation and Commitment Account (RQ3)

The modes show what happens, and the account asks why the work takes this shape. Reading the 38 delegation episodes against the codes suggests a cycle of three parts. The human delegates a scope, the AI generates within it, and the human commits to or reformulates what the AI returns. Figure 3 shows the cycle with one message from the project for each step. Three mechanisms produce it, and Table 3 states each as a proposition with its evidence and the evidence that would count against it.

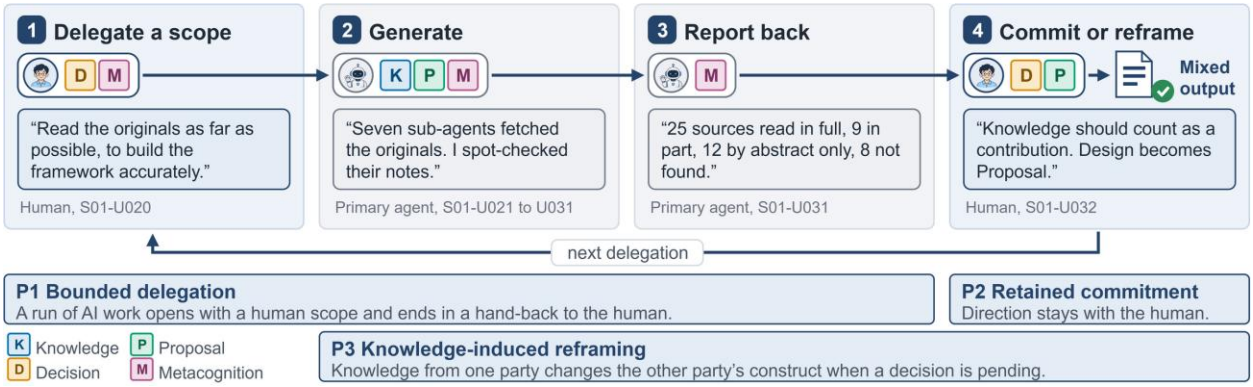


Figure 3. The cycle of delegation, generation and commitment, with the three propositions and one message of the project for each step. Message texts abridge the project messages.

The first mechanism is bounded delegation. Human Decisions and Metacognition open runs of AI work. The median run holds 2.5 AI responses. Eleven of the 38 runs hold 120 of the 177 primary-agent responses, and 113 (GPT) and 110 (Gemini) of these 120 carry Execution or Metacognition as the primary tag. Each long run began with a human message that fixed a multi-step piece of work. The instruction to read originals (S01-U020) opened 11 responses of source retrieval and checking, which ended in a synthesis (S01-U031). The instruction to dispatch the coders and, in our translation, "write the complete one" (S01-U045) opened 18 responses. They ended when the human stopped the AI from drafting the paper (S01-U064), because the AI had read the phrase as an instruction to write the paper (S01-U065).

Nine long runs were followed by a human message. Five ended with a report of more than 1,000 characters that the human answered. Four ended while the AI was working in short steps, and in each the human stopped, rejected, corrected or extended the work (S01-U064, S01-U107, S01-U126, S01-U164). A run therefore ends with a hand-back to the human.

The second mechanism is retained commitment. Decision appears in 28 of 38 human units and in 4 of 187 primary-agent units. Three of the four AI Decisions assign tasks to sub-agents inside a delegation. The fourth is the AI's own decision not to recode after the framing revision (S01-U185), and the human overturned it in the next message. The human message that follows an AI Proposal carries a Decision in 19 (GPT) and 18 (Gemini) of 21 cases, more often than the 74% for human messages in general. Commitment therefore belongs to the human role whatever the AI proposed. All three human Proposals came directly after an AI unit that carried Knowledge or a Proposal, and two reformulated what the AI had returned (S01-U032, S01-U070). The human stated at the start that execution follows a blueprint that the human draws (S01-U005), and the project rules, drawn from the call, state that humans orchestrate and remain accountable. Baird and Maruping (2021) note that agentic artifacts can assume responsibility for tasks. Here responsibility for tasks moved to the AI while accountability for the direction stayed with the human.

The third mechanism is knowledge-induced reframing. AI-supplied knowledge changed a human construct twice. After the synthesis of the readings (S01-U031), the human stated that the content on cognition and

its externalised forms had been unknown before and made knowledge a type of contribution (S01-U032). The AI then added a flag for knowledge transfer to the framework (S01-U034). After the AI explained agreement coefficients (S01-U044), the human decided to report several of them (S01-U045). In both cases a decision was pending and the human lacked a piece of knowledge that bore on it. The mirror case also occurs. The AI had reported that image generation was unavailable, the human supplied situated knowledge about an existing configuration (S13-U006, S13-U010), and the AI changed its route. Supply and reframing is rare among the transitions, yet each instance changed the design.

The three mechanisms together produce mixed contributions. A run delivers Knowledge or a Proposal, and the human's commitment adds content in the form of a reframed construct, a replaced candidate set or a new rule. Each turn builds on the output that the other party left, which resembles the imbrication of human and material agencies that Leonardi (2011) describes. Where the commitment adds nothing, as in the short approvals S01-U078 and S01-U080, the output stays with one party.

Proposition	Evidence in the case	Evidence that would count against it
P1 Bounded delegation. A run of AI work opens with a human scope and ends in a hand-back to the human.	11 of 38 runs hold 120 of 177 primary-agent responses, and 113 (GPT) and 110 (Gemini) of those are Execution or Metacognition. Of 9 long runs followed by a human message, 5 ended with a report and 4 with an intervention.	Long runs that end with neither a report nor an intervention.
P2 Retained commitment. Responsibility for tasks can move to the AI while accountability for the direction stays with the human.	Decision in 28 of 38 human units and 4 of 187 primary-agent units. After an AI Proposal the human message carries a Decision in 19 and 18 of 21 cases.	AI Decisions that set goals or scope and stand without human adoption.
P3 Knowledge-induced reframing. Knowledge from one party changes the other party's construct or route when it fills a gap behind a pending decision.	AI to human: S01-U031 to S01-U034 and S01-U044 to S01-U045. Human to AI: S13-U006 and S13-U010.	Knowledge supply that leaves the construct or route unchanged although a decision is pending.

Table 3. Propositions of the delegation and commitment account with evidence from the case.

Discussion

Theoretical Contributions

In Gregor's (2006) taxonomy, the typology is a theory for analysing, of the kind she calls a classification schema. The three propositions are a step toward a theory for explaining, which she describes as accounting for how and why events occur and which can take a process form. The account extends the delegation lens (Baird & Maruping, 2021). Delegation describes what is handed over, and the account adds what thinking comes back and which party holds the commitment. It also locates a different problem from the one that Fügener et al. (2022) study. Their humans misjudge what to delegate to AI, whereas the failures here lay in a wide reading of an ambiguous instruction (S01-U064) and in reports that the human found inaccurate (S01-U126, S01-U134).

Treating Knowledge as a contribution type adds a direction to the human-in-the-loop picture. In the configurations that Grønsund and Aanestad (2020) describe, humans audit and alter the algorithm. Here the AI also changed the human's constructs, in at least two episodes. The account also raises a question for the performance literature. Combinations lose in decision tasks and gain in content creation (Vaccaro et al., 2024), and in this case the human kept the decisions while the AI produced much of the content. Whether retained commitment explains or counteracts the reported loss is open.

Implications for Practice and Governance

Journals, conferences and workshops that require disclosure can ask for more than a task list. A disclosure can state who held the commitments, which runs of work proceeded between human messages and what scope bounded them, and which outputs combined contributions from both parties. Organisations that

deploy AI agents can treat the scope statement and the closing report as control points, since the overreach in this case followed an ambiguous scope and two reports were inaccurate. The proposition on retained commitment gives accountability a concrete reading, since the human can be asked to answer for every Decision and every adopted Proposal in the record. The recoding of an extended corpus left the agreed presence of actor-type pairs nearly unchanged, so the standard can be applied again at each revision. Any project that keeps its sessions can produce such an account.

Limitations and Future Research

The coding standard performed unevenly, and the unevenness is informative. Kappa between the two coders ranged from .22 to 1.00 across the actor-type presence variables, and alpha stayed within .04 of it (Appendix Table A8). Human Knowledge (1.00), Metacognition (.92) and Decision (.91) and primary-agent Metacognition (.91) were highest. Primary-agent Decision (.22) and Knowledge (.44) were lowest, because GPT tags Knowledge in 99 of the 187 primary-agent units and Gemini in 42, and Decision in 23 and 8. The coders agreed on the primary type in 226 of 265 units (85%), and the main disagreement lay between Metacognition and Execution (13 units), followed by Metacognition and Proposal (9). On the 146 units that both runs coded, each coder repeated its own primary type in 90% (GPT) and 95% (Gemini) of cases and its full set of types in 79%. Agreed presence of an actor-type pair changed by at most three units, and no pair changed between present and absent. Kappa between the coders rose for eight of 13 variables, stayed equal for two and fell for three, for instance from .57 to .26 for primary-agent Decision (Appendix Table A11). Conclusions therefore rest on agreed presence and not on one coder or one run.

The study has limits that bound its claims. It covers one project with one human orchestrator, and the propositions await tests in projects with several humans, other agent systems and other tasks. The codebook treats an AI draft as a Proposal until a human adopts it, so part of the contrast between AI Proposals and human Decisions follows from the definition. The primary agent, the sub-agents and the author of the typology share a model family that differs from both coders. This reduces but does not remove the self-preference that LLM evaluators show toward their own generations (Panickssery et al., 2024). Two LLM coders are fewer than the three or more that reliability guidance recommends, and their runs are not independent coders in the strict sense (Artstein & Poesio, 2008). An LLM agrees with experts less than experts agree with each other (Xiao et al., 2023), and validation against human labels is recommended (Pangakis et al., 2023, Ziems et al., 2024). The typology was developed on the early part of the corpus, and 129 of the 265 units were produced after the codebook was frozen, although by the same agent.

Cognition that never entered the trace, such as reading, reflection and conversation with the advisor, is invisible to the method. The primary agent drafted the qualitative interpretations, including the mechanisms. The propositions were built from the episodes of one project and remain conjectures until other projects test them. The human has reviewed the definitions, the example cases and the propositions, and not every cited unit. Some sources were read as preprints or third-party copies. Costa et al. (2025) and most IS works in the background were read as abstracts and records, and only Gregor (2006) in full, so claims of novelty rest on a limited literature search. Future work can test the propositions in projects with several human authors, add human coders for a validation subsample and repeat the recoding for each further revision.

Conclusion

In this project the thinking was shared through a cycle of delegation, generation and commitment. The human delegated scopes and committed to the direction of the work. The AI generated knowledge, proposals and verification inside the scopes, and mixed contributions arose where the human's commitment added content. The cycle answers the two worries with evidence. The human remained necessary because commitment stayed with the human, and the AI did more than execute because it supplied the knowledge and proposals on which commitments were made. The typology and the three propositions give disclosure and governance a vocabulary beyond task lists and give IS research claims to test. A disclosure that follows the account states who delegated, who committed and what each party added to a shared output. Because this paper is part of the data it analyses, the sequence of its revisions belongs to the method.

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Workflow Appendix: How a Human Orchestrator and AI Agents Produced This Paper

This appendix documents the research workflow of the submission. It follows the items required by the call, namely a chronological map, models and tools, representative prompts, errors and abandoned approaches, material human interventions, verification steps, artifact availability and a contribution account. The complete logs, codebooks, scripts and figure assets are in the artifact bundle. The raw session exports are withheld for the reasons given in Section G. Identifying names, credentials and local paths are removed in this document. All text of the paper and of this appendix and every figure was produced by AI agents. The human typed no manuscript text and gave instructions in conversation.

A. Chronological Workflow Map

Stage	What happened	Main actors
1 Ideation	The human forwarded the workshop call, proposed the topic of cognitive contribution distribution and asked for an assessment. The AI judged fit, listed risks and suggested a process-log method. The human decided to open a dedicated project (I004, I005).	Human, AI
2 Task framing and recording	The AI read the call files and summarised duties and tensions. The human fixed scope, self-reference, project-level recording of every session, verified references and page limits. The AI built the recording mechanism, project rules and a project skill (I002, I003, I007).	Human, AI
3 Literature survey	The human rejected the AI's memory-based taxonomy and demanded a survey (I010). Three sub-agents surveyed cognition, human-AI contribution and coding methods, and returned 75 candidate sources with bibliographic verification.	Human, AI, sub-agents
4 Full-text reading	The human asked for original texts (I011). Seven sub-agents read open versions and wrote notes with access levels. Two crashed on a PDF error and were rerun. Unreachable closed sources were skipped by human decision (I012).	Human, AI, sub-agents
5 Framework and codebook	The human corrected the AI's treatment of knowledge as a dimension and set four types (I012). The AI drafted framework v0.1 and v0.2, then codebook v1.0 with a response-level unit and multi-label tags. The human ruled on boundary cases and fixed the unit (I013, I014, I015).	Human, AI
6 First coding run	The AI built coding packets, and GPT-6.1-sol and Gemini-4-argon coded 106 units independently under codebook v1.0. Agreement was computed with several statistics.	AI, coders
7 Research questions and v1.1	The human set qualitative research questions and the definition of mixed contributions, and ruled on example cases (I017 to I021). The AI drafted codebook v1.1, which the human approved (I022). The same units were recoded and compared.	Human, AI, coders
8 Full coding and analysis	The corpus was extended and coded under v1.1 (146 units). The AI carried out the qualitative analysis of existence, distribution and transitions.	AI, coders
9 Figures and manuscript	The AI drew flat figures that the human rejected (I024) and then redrew them with the Sivia workflow. The human approved whole-figure drafts (I029) and asked for concrete examples in the speech bubbles (I030). A second agent session on another model redrew both figures and the human rated the result as poor (I034 to I036). That session finished Figure 2 after a model change, and Figure 1 was completed in the main session (I037, I038). The AI drafted the manuscript under the human-style writing constraints, and this appendix.	Human, AI, image model
10 Audit and repair of the record	While auditing the second session, the primary agent found that its own run of the sync script had replaced 29 raw session files with help text. It rebuilt every session file from the OpenCode database and rewrote the script (E057, E058). The account is in Section E.	Human, AI

Stage	What happened	Main actors
11 Structure and layout review	The human noticed that the paper had no abstract, that the Introduction was short and that some pages ended in white space (I040). The AI measured the section proportions of 16 ICIS papers and found that the template keeps the abstract out of the paper file. It rewrote the Introduction with concrete content, added eight verified sources and a coding excerpt, repaired a citation that was missing from the reference list and removed the gaps (E063 to E067).	Human, AI
12 Framing revision	On 26 October a co-author judged that the paper did not read as an information systems paper, and the human added three problems with the research questions (I041, I042). The human chose to restructure it as a conceptual paper positioned in IS theory. The AI set off plain research questions, added an IS background from nine verified sources, derived delegation episodes and the fate of AI proposals from the coded units, and wrote three propositions. The human then rejected the shaded question box and the grid tables, asked for the original title and wording to be kept, and asked for the figures to fill the pages (I043 to I045, E075). While editing one label of Figure 1 the AI overwrote the final figures with an earlier design and restored them (E073).	Human, AI
13 Recode and layout	Following the human rule and request, both coders recoded the 146 earlier units and 119 later ones under codebook v1.1 (run 2, E076). Run 2 replaced run 1 in all counts, and Table A11 compares the runs. The AI rebuilt the mechanism analysis on the extended corpus, built Figure 3, rebuilt Figure 2 after finding that it still showed the counts of run 1 (E080), cut about 600 words to bring the body back to 8 pages, moved Figure 1 into Research Design and compacted the tables. The human dropped the Use of Generative AI section because the submission form has the item, after the AI had checked the call for papers (I045, I046, E078, E079).	Human, AI, coders
14 Submission package	The AI assembled the single PDF, drafted the text of every form field and built a redacted artifact bundle (I047). A scan of the appendix found garbled rows, resource names that contain the author's first name and a prompt description in the imperative, and these were repaired. The human asked that the verbatim originals be kept, and the Chinese wording of seven messages was restored from the raw sessions (I048, E082, E083). The raw session exports are withheld because a pattern-based redaction cannot make them safe with certainty (E084).	Human, AI

Table A1. Chronological map of the project.

B. Models, Agents, Tools and Environment

Role	System	Notes
Primary agent	claude-sonnet-5.5 (agent name openwork)	Run in OpenCode 1.18.30 through OpenWork 0.18.59. Sampling settings are not exposed to the agent. Model, agent name and system prompt of every message are in the raw sessions.
Sub-agents	claude-sonnet-5.5 (general sub-agent type)	Literature survey and full-text reading. Each is a child session of the main session. Prompts required compliance with the project rules.
Independent coders	GPT-6.1-sol, Gemini-4-argon	Run through the same agent framework in isolated directories that held only the codebook and unit packets. Chunks of up to 20 units. Run 2 used four parallel jobs of 136 units, two per model.
Image model	Azure OpenAI image deployment (gpt-image-2.5 family)	Three icon sheets at 1536 by 1024, high quality. Each call is recorded with prompt and hash.
Bibliographic services	Crossref, OpenAlex, DataCite, arXiv, OpenLibrary	Used for existence checks and open-access location. Access was limited to legal sources.
Local tools	Python 3.11 (pypdf, Pillow, numpy, matplotlib, python-docx), draw.io desktop, Microsoft Word, Windows PowerShell 5.1	Windows 10.0.26300. Word was used only to count pages.

Role	System	Notes
Figure model	opencode big-pickle and claude-sonnet-5.5	A second agent session redrew the figures. The first part of it (intermediate versions of Figures 1 and 2) ran on big-pickle. After the human rated the aesthetics as poor (I036) the session continued on claude-sonnet-5.5, which redrew Figure 2. Figure 1 was redrawn in the main session with the same drawing library. Some log rows call big-pickle MiniMax-Flash, which the session records do not confirm.
Figure method	Sivia workflow: full prompts checked against the template length, whole-figure raster drafts from an Azure OpenAI image model and human approval, then 30 icon assets with key-colour cut-out and quality checks. The final figures are drawn through the live draw.io tools and checked by the live audit, a geometry check, a blank-area check, and renders viewed in colour, in grey and at print size. Figure 3 was built the same way and audited with 0 hard failures and 0 warnings (V061).	The first editable figures were generated as XML files by a Python script because the live tools were not available, and an earlier statement that the Sivia workflow had been used for them was wrong. The human rejected those figures and two later versions. The image call was ported to Python because the original script needs Node.

Table A2. Models, agents and tools.

C. Representative Prompts and Instructions

Human instructions were given in Chinese and are paraphrased here. The first project instruction asked the AI to read the call and to set the deliverables, namely a paper, a theory-grounded coding standard, a provenance-tracing method and skills. It also asked for a full record of the collaboration sessions. A decisive redirect asked that the externalised forms of cognition be modelled from existing research and not guessed in conversation. Later instructions fixed the coding unit as one model response and named the two coder models. They defined mixed contributions by explicit expression from both actors and required that writing and figures follow human-style constraints. The last redirects asked that the research questions stand out and read plainly, that they gain theoretical abstraction, and that the findings explain mechanisms and not only count (I042). Later instructions asked for a plain bullet list instead of a shaded box, three-line tables, a recode, and figures arranged so that no page keeps a large blank area (I043, I044).

Prompts that the AI wrote to sub-agents began with the project rules and a research brief. The literature prompts asked for verified sources only and told the agent to mark unverified items. The reading prompts required notes with an access level for each source and short quotations with locators. They also required a statement that a source was not accessed instead of filling gaps from memory. The coder prompt told each coder to read the coding manual and apply it to every unit of the packet. It also told each coder to code only what is observable and to take the actor hint as the actor of the tags. Each coder was to write one JSON line per unit in the schema of the manual and to read no other files.

D. Material Human Interventions

ID	Stage	Type	Intervention and reason
I001	problem_formulation	topic_selection	Human proposed the paper idea: cognitive-contribution distribution in human-AI research collaboration, for ICIS 2026 PDW on Autonomous Scholarship (details in origin session, read raw file)
I002	problem_formulation	scope_and_requirements	Human specified deliverables: paper + cognitive-contribution coding standard with classification rationale (theory-grounded) + provenance-tracing method + skills, require full recording of collaboration sessions for later disclosure, asked AI to first read the call for papers
I003	governance	process_rules	Human decided: (a) self-referential design is intended, this project's own sessions are the sole data, (b) execution is AI's job but contains little cognition (blueprint-following), (c) recording must be project-level mandatory, covering all sessions incl. other agents/new sessions/sub-agents, with model, agent framework and ...

ID	Stage	Type	Intervention and reason
I004	problem_formulation	topic_selection	Human forwarded the ICIS PDW call-for-participation email (sent by organizer an organiser via AISWorld) and proposed the topic: cognitive contribution distribution in human-AI collaboration, asked AI for its view
I005	governance	process_design	Human acknowledged breadth, decided to open a dedicated project with an Agent to converge topic, design and coding method, and to disclose method and interaction process
I006	governance	task_assignment	Message 1 of this session (retroactive entry, see I002): asked AI to read the CFP and understand the task, set deliverables and recording requirement
I007	governance	process_rules	Message 2 of this session (retroactive entry, see I003): governance decisions and instruction to build the recording mechanism first, then start research discussion
I008	governance	process_rules	Human: routine notices need no reply, end-of-project full recording+disclosure is what matters, raw data must be anonymised, submit and open-source de-identified versions, AGENTS.md content judged acceptable (E005 closed), advisor is NOT an organiser (no organizer conflict)
I009	theory	construct_proposal	Human proposed to first model cognition and especially its externally observable forms, then model interaction behaviour, folk-theory proposal that cognitive contribution has three types: decision, knowledge, expression/design
I010	literature	redirect	Human: yes, cognition AS EXTERNALLY OBSERVABLE FORM is what must be modelled, this cannot be guessed by human and AI in conversation, so AI should first survey related research
I011	literature	quality_requirement	Human: read the ORIGINAL texts as far as possible to build the framework more accurately
I012	theory	construct_revision	Human: unavailable (non-open-access) sources can be skipped. Human challenged AI's reading of knowledge as a mere dimension: in the current paradigm agents supply NEW knowledge to human experts (e.g. human learned the cognition literature only after the AI summarised it), so knowledge should be a contribution type, ...
I013	design	codebook_decisions	Human decisions on framework v0.1: (1) evaluation and choice do differ conceptually but are hard to separate, evaluation usually carries preference/choice, (2) coding unit = one round, a round may mix several types and even human and AI contributions at once, (3) human will bring in two new models as coders, AI acts as main ...
I014	design	unit_clarification	Human corrects the AI's reading of 'round': a round is ONE model response, not human-message-to-human-message (that would be too long), a sub-agent dispatch counts as a round, possibly several. Human will use GPT and Gemini as the two coders, wants the whole paper written TODAY and the codebook fixed soon, agrees to a dry run
I015	design	codebook_approval	Human: report several agreement statistics incl. kappa, no threshold for now (small sample, mainly qualitative, decide after results), unit definition (text response + preceding tool steps) OK, v1.0 to be frozen on this confirmation, AI should dispatch opencode sub-agents with models gpt-6.1-sol and Gemini 4 as coders, write ...
I016	governance	stop_and_redirect	Human: no rush to write the paper, the complete framework and the logical structure are not designed yet
I017	problem_formulation	research_question_direction	Human: next consider research questions, beyond the basic cognitive taxonomy and coding standard, since only this project's content is analysable and it is small, prefer a qualitative study rather than quantitative
I018	problem_formulation	research_questions	Human sets 3 RQs for a qualitative study (focus on existence and patterns, not point-to-point relations or numeric regularities, distribution only symbolically): RQ1 existence of human, AI and mixed human-AI cognitive contributions (answers 'does AI make humans useless?' and 'is AI only an execution tool with no ideas?', ...

ID	Stage	Type	Intervention and reason
I019	design	mixed_contribution_definition	Human: a mixed human-AI contribution = both actors contribute to the SAME output (e.g. human proposes a scheme drawing on AI-organised knowledge: AI knowledge + human proposal = mixed). It must be an explicit expression: a run of bare 'confirm'/'OK' approvals does NOT count (guards against rubber-stamping). Transitions are ...
I020	design	example_rulings	Human reviewed extracted examples: (1) examples 1-2 correctly illustrate mixed contribution, (2) human correction (corrective redirect) is NOT a separate transition category - it shows up normally in the next round, (3) AI narration of its own next steps counts as execution (X) with no cognitive contribution
I021	design	research_design_answers	Human: (1) output-level identification of mixed contributions is fine, (2) RQ2 distribution should also be reported by research stage, (3) no separate section on AI autonomous decisions needed
I022	design	codebook_approval	Human approves ("可以") freezing codebook v1.1 (C1-C7) and re-coding the same units
I023	governance	go_ahead	Human: from the example review, go straight to FULL coding, then qualitative analysis, then complete the paper. Writing and figure drawing must follow the human-style writing constraints, figures may use the Sivia method ([local path]). Sub-agent M problem: keep as a Discussion-level limitation, no v1.2
I024	revision	figure_quality_feedback	Human: the figures are too plain ('low'), use image generation to make icons, assemble them in draw.io, and give them a refined look, supplied a reference overview figure (thesis-style multi-panel figure with coloured chapter bars, small illustrated icons, labelled arrows, legend strip)
I025	writing	layout_requirement	Human: the layout is abnormal, the main paper should be exactly 8 pages
I026	revision	process_challenge	Human asks whether Sivia was really used, since the figure style differs greatly from the reference and the information density is low
I027	revision	figure_redo_decision	Human: redo the figures completely with the full Sivia workflow, change figures first, page count later, figures in English, the earlier reference image was only for reference and is dropped if Sivia is used fully
I028	revision	figure_title_removal	Human: no title inside the figures, the captions carry the title
I029	revision	visual_approval_and_reconstruction	Human: density of the drafts is acceptable, start the editable reconstruction, the weaknesses AI listed must be fixed, English labels are fine
I030	revision	figure_content_density	Human: put concrete examples inside the speech bubbles to raise information density and use space better, since some areas are still empty
I031	revision	figure_redraw_request	Human: "帮我重新绘制一下两个图，用 Sivia 来画，可以生成其中的元素，遵循人类风格的约束" (redraw the two figures with Sivia, generated elements allowed, human-style constraints)
I032	revision	tool_availability_question	Human: "Azure 那个绘图技能不能用吗，我记得已经配置了呀" (the Azure drawing skill cannot be used, it was configured already)
I033	tooling	tool_availability_question	Human: "[resource]下边有个 2.5 吧我记得" (there is a 2.5 under [resource], I remember)
I034	figures	figure_review_feedback	Human review of the delivered Figure 1 PNG: "我觉得对话框的形状有点奇怪，箭头起码应该是在侧面而不是下面，而且人类和 agent 最好分列两侧。Tags 这个 panel 里面，有比较多的空白区域" (the dialogue box shape looks odd, the arrow should at least be on the side rather than below, human and agent should be separated on two sides, the Tags panel has a lot of blank space)
I035	figures	figure_review_feedback	Human review of the first v2 draft: "对话的部分，分列两侧不是说两边分庭抗礼，是可以堆叠的（但不要重叠），现在这样又出现了很多空白区域" (in the dialogue part, putting the two sides apart does not mean the two sides should face each other like two armies, the messages can be stacked on the same side but must not overlap, and now there are large blank areas again)

ID	Stage	Type	Intervention and reason
I036	figures	figure_review_feedback	Human review of the delivered Figure 1 v2 and Figure 2 PNGs, after the model was switched: “重新再画一下，刚才那个模型审美有点次。” (redraw it again, the aesthetics of the previous model were a bit poor)
I037	revision	figure_integration_and_audit	Human: I redrew the two figures in another session, please update the paper and also check what that session reported
I038	revision	decision_between_options	Human, asked where the new Figure 1 was (it did not exist, the other session stopped after Figure 2): let the AI draw Figure 1
I039	revision	figure_consistency_and_disclosure	Human: (1) the worked case in Figure 2 must agree with the text, (2) the record incident may be written into the appendix, and the human considers it a valuable record
I040	revision	structure_and_layout_review	Human: (1) the paper has no abstract and the Introduction is short, check the structure and the length distribution against ICIS papers accepted in earlier years. (2) Pages 3, 5 and 6 end in about 5 to 8 lines of white space, fill them with concrete content, not by lengthening sentences, and follow the human-style constraints
I041	writing	external_feedback_and_question	Human relays co-author feedback: the paper does not look like an IS paper, 'it would be state the research question clearly and write it like a traditional information systems paper'. The human asks how the current paper differs from a traditional IS paper
I042	revision	theory_and_framing_redirect	Human adds three problems to the co-author's feedback: (1) the research questions are not visually prominent and sit inside a paragraph, (2) they stay at the level of statistics of one concrete project and lack theoretical abstraction, so they should be intuitive and plain first and return to formal definitions and counts when ...
I043	revision	style_scope_and_decisions	Human reviews the revised draft. (1) No shaded box for the research questions, use a normal paragraph and a bullet list. (2) Tables should be three-line tables or a derivative of them, as in ordinary papers. (3) Asks why the research became 'AI-led' and says that when something is flagged the AI should change only that. ...
I044	figures	layout_feedback	Human on the layout: do not leave such large blank areas. If the size is hard to control, the figure may be set with text wrapping
I045	writing	structure_decision	Human on layout: figures that do not fit should be moved or re-arranged so that pages are filled, not left with large blank areas. Human then decides that the Use of Generative AI section does not need to be in the paper because the submission form has this item
I046	writing	requirement_confirmation	Human asked the AI to check the CFP for a requirement of a Use of Generative AI statement in the paper, and then confirmed that the submission form has this item, so it need not be written in the paper. The human also said that the less trimmed version would then be enough
I047	submission	task_assignment	Human asks the AI to prepare the submission materials according to the OpenReview submission form (single PDF with main paper, references and workflow appendix, and the disclosure fields: title, keywords, TL,DR, abstract, AI systems and tools, nine human-AI contribution selections, summary of material human interventions, ...
I048	submission	record_requirement	Human on the cleaning of the appendix: the original verbatim versions must also be kept
I049	submission	disclosure_correction_and_decisions	Human states that the Writing stage should be 'AI only' because the human has not typed a single word of the manuscript. Human created a GitHub repository for the artifacts and asks the AI to upload the bundle later, and will paste an anonymous GitHub link into the form. The forwarded call-for-participation email must not be ...
I050	submission	repository_history_request	Human clarifies that the submission is made under the human's own identity and that the pasted link will be an anonymous GitHub mirror, not the repository itself, then asks the AI to refresh the commit history of the repository

Table A3. Human interventions recorded during the project.

E. Errors, Abandoned Approaches and Responses

ID	Type	Event and response
E001	tool_failure	webfetch of ICIS CFP URL (guessed URL) failed with transport error Response: Fell back to CFP files already in workspace/CFP, no content lost
E002	environment_limitation	python, node, git, sqlite3 not on PATH in the agent shell, PATH lacks System32 Response: Implemented recording tooling in PowerShell (full path) + opencode CLI export instead
E003	log_correction	I001 was written before the AI had read the origin session and its summary is vague, I004/I005 are the faithful records and supersede it. I006/I007 attach exact message ids to I002/I003 (retroactive) Response: Appended corrective rows (append-only rule)
E004	tool_failure	sync script first run failed: PATH lacked System32 (cmd not found), then non-UTF8 decoding of opencode session list output broke JSON parsing Response: Script now prepends System32 to PATH and reads CLI output via UTF-8 temp file, re-run OK, 2 sessions captured
E005	file_overwrite_notice	AGENTS.md was already loaded in the system prompt but absent from an earlier dir listing, AI wrote AGENTS.md without reading it first. Written content mirrors the rules loaded in the prompt Response: Ask human to confirm AGENTS.md content
E006	tool_failure	First attempt to append E003-E005/I004-I007 and set roles via inline cmd->PowerShell quoting silently failed (row counts unchanged) Response: Re-done with a script file
E007	event_closure	E005 (AGENTS.md written without prior read) closed: human reviewed and accepted content (see I008) Response: none needed
E008	abandoned_approach	AI proposed a function x stage x attribution taxonomy from its own memory (no sources) in the previous turn, human redirected to literature-first (I010). Proposal kept only as a hypothesis to test against the literature Response: Literature survey commissioned
E009	subagent_spawn	Spawned 3 sub-agents (type general) in parallel: A=cognition forms/taxonomies (task ses_ea307c02bffeHFBzAFG8xzW7wK), B=human-AI contribution/IS/GenAI-in-research (task ses_ea3079637ffegcYEhVlpvXOQLD), C=coding schemes/LLM-coder methods (task ses_ea3075e9dffepzhhWfOy15F93b). ... Response: All 3 returned
E010	tool_failure	Crossref/arXiv/Semantic Scholar returned HTTP 429 repeatedly under parallel load, Semantic Scholar returned null abstracts, Crossref lacks books (verified via OpenLibrary), some Crossref records incomplete (Nelson and Narens, Rai et al. Editors Comments, Hadwin year) Response: Serialised requests, flagged items. Consequence: for most sources only the bibliographic record was verified, content NOT read
E011	error_caught	Brief given to sub-agent had errors that sub-agent C corrected against records: Amershal->Amershi, CoAIcoder is ACM TOCHI 2023 not CSCW 2024, Hadwin et al. year missing in Crossref Response: Corrected in sources_candidates.csv
E012	tool_limitation	opencode session list hides child (sub-agent) sessions, sync script initially missed the 3 literature sub-agent sessions (parentID set, same directory, exportable by id) Response: sync_sessions.ps1 now scans exported transcripts for session ids and adds those with parentID, 3 child sessions captured with parent link
E013	coding_caveat	In child sessions the single role=user message is the PARENT AI's prompt, not a human message. Any coder or script that infers actor from role=user will mis-attribute: actor must be derived from session lineage (parent_session_id) and message source, not role Response: Record in codebook (unit-of-analysis / actor rules) when it is written
E014	subagent_spawn	Round 2 full-text reading: spawned 4 sub-agents (type general) G1 cognition (ses_ea2fdf10affejkfzAd0fMZsLJI), G2 coding schemes (task ses_ea2fdcef5ffeXKndXENbHHYRxK), G3 human-AI (ses_ea2fda8d8ffeZDcicyKTJC6TRa), G4 methods (task ses_ea2fd860cffeg2KP3aSGwTejfk). Prompts required ... Response: G1 and G3 completed, G2 and G4 crashed
E015	subagent_failure	G2 and G4 sub-agents failed with 'invalid PDF data: missing PDF header' (a non-PDF response was attached as PDF). G4 had already written 7 notes (S062 S065 S066 S067 S068 S070 S071) before crashing, G2 wrote none Response: Re-spawned as G2a G2b G4b with instruction to use webfetch format=text only and never attach web PDFs, parent spot-checked S068 and S071 content as ...
E016	subagent_spawn	Re-spawned: G2a (ses_ea2e936ffffeS9S8yrARqHuQsx), G2b (ses_ea2e91aa9ffeDU95gks8Fcqxu0), G4b (ses_ea2e8f9caffe5i41TQ760735pH) Response: All 3 returned, G2a read nothing (all NOT_ACCESSED), G2b read 2 fully, G4b read 5 fully

ID	Type	Event and response
E017	self_correction	G3 sub-agent reported it first wrote author lists and volume/issue numbers from memory into notes S043 S030 S042 S052, then caught and removed them on review Response: Removed, those notes contain only fields it saw. Parent has not independently re-read S043 S030 S042 S052 so bibliographic fields there remain ...
E018	security_deviation	G1 sub-agent disabled TLS certificate validation once in a download attempt (johngero.com), the attempt failed anyway. The reader tool also auto-copied S014 and S017 PDFs plus rendered page images into .opencode/openwork/inbox/ inside the workspace Response: No data obtained through the TLS bypass. Inbox files are publisher/author PDFs of copyrighted articles: exclude from any public repository (add to ...
E019	provenance_caveat	Several read copies are third-party hosted, scans or preprints: S007 (DLI OCR scan of 1960 book, rights unclear), S008 (reprint, different pagination), S072 (third-party JSTOR scan), S076 (SciSpace copy), S061 (author manuscript), S029 (manuscript), S013/S002/S017 ... Response: Quotes and page numbers must be re-checked against licensed versions before manuscript, do not redistribute copies
E020	access_failure	Key sources could not be read legally by script: Chi&Wylie, Chi 1997, Ericsson&Simon, Clark&Brennan, Goldschmidt, Hesse, Campbell, Gregor&Hevner, Dell'Acqua (closed), Klahr&Dunbar, Dorst&Cross (403/abstract only), Costa 2025 (CC-BY but SAGE 403/Cloudflare), Weinberger, ENA ... Response: Ask human to supply PDFs via institutional access (legitimate human action to be logged), no workaround to bypass access controls
E021	tool_discovery	Python 3.11 with pypdf exists at %LOCALAPPDATA%\Programs\Python\Python311\python.exe (not on PATH), parent created research/tools/fetch_pdf_text.py and used it to read S029 (Garrison) from an author-hosted Wayback PDF Response: environment.md to be updated
E022	ai_framing_error	AI's evidence synthesis v1 treated knowledge only as a dimension crossing cognitive processes (following Bloom revised/de Jong), which silently assumes the contributor already holds the knowledge. Human pointed out this assumption breaks when AI supplies new knowledge to the ... Response: Knowledge reinstated as a contribution type with an epistemic-transfer flag in framework v0.1
E023	ai_misreading	AI interpreted human 'one round' (I013) as human-message-to-human-message and warned of only ~6 rounds, human meant one model response. Misreading cost a wrong small-N warning in the previous turn Response: Unit redefined, N estimate recomputed from data
E024	codebook_freeze	CODEBOOK v1.0 FROZEN. All units existing at this timestamp form the DEVELOPMENT set, units created later form the EVALUATION set. Any change creates v1.1 and is logged. Sha256 of frozen file recorded in verification.csv
E025	coder_dispatch	Coders dispatched by AI via 'opencode run -m' (task tool cannot select models): coder A = github-copilot/gpt-6.1-sol, coder B = github-copilot/gemini-4-argon (the only gemini-4 id available, human wrote 'Gemini4' so this mapping is an AI assumption). Each runs in a neutral temp ...
E026	tool_failure	Attempt 1 (whole packet, one run each) failed: GPT run ended with 'unknown certificate verification error' after ~18 min with no output, Gemini ended with finish reason 'length' (output too long) without writing result file. Logs kept in temp as failed_attempt1_*.json Response: Re-run in chunks of 18 units with retry (research/tools/run_coder.ps1)
E027	ai_overreach	AI read 'you can dispatch the coders and write the complete one' (I015) as 'write the full paper' and began generating a full manuscript draft (research/tools/build_paper.py, paper/ not yet produced because the docx build failed on a missing 'Table Grid' style) before the ... Response: Paper writing stopped. build_paper.py kept as a record of the abandoned premature draft, its text must not be reused without redesign
E028	coding_run_completed	Development-set coding run (codebook v1.0, 106 units) finished: GPT-6.1-sol and Gemini-4-argon each returned 6 chunks, all units coded. Outputs: research/coding/agreement_by_actor_type.csv, presence_by_actor_type.csv, confusion_primary.csv, fig_presence.png. Observations bearing ... Response: Findings treated as development-set, exploratory, used to inform codebook v1.1 design, not as results yet
E029	filter_weakness	examples.py mixed-candidate filter is crude: length and tag-based, not content-based. It lets through cross-session pairs (S02-U007->S01-U001) and wrongly drops short substantive instructions (S01-U020 'read the originals'). Also AI 'narration' units ('first I will log this...') ... Response: Present as examples only, rubber-stamp test must be content-based, decide narration coding rule with human

ID	Type	Event and response
E030	codebook_freeze	CODEBOOK v1.1 FROZEN (changes C1-C7). v1.0 coding results remain as run 1 (same 106 units, development set). Run 2 re-codes the same units under v1.1, NOTE the v1.0 and v1.1 runs share the same units, so run 2 is still development data, not an evaluation set
E031	recoding_run_completed	v1.1 re-coding of the same 106 units finished (GPT-6.1-sol, Gemini-4-argon, 12 new coder sessions registered). Outputs research/coding/agreement_by_actor_type11.csv, confusion_primary11.csv. Compared with v1.0: X agreement for primary AI rose kappa .35->.87 and sub-agent ... Response: Report as version comparison (research design step 5), note sub-agent M disagreement as a boundary problem for v1.2 discussion
E032	quality_rejection	First figure versions (flat draw.io boxes and a numeric matrix) were rejected by the human as low quality, AI had judged them clean after its own visual review, so its review criterion (alignment, legibility) missed the richness expected for a publication figure Response: Figures to be rebuilt with generated icons, earlier versions kept as fig*_v0 records
E033	tool_adaptation	Sivia generate-figure-elements pipeline requires node (imagegen.mjs), node is not installed. AI wrote a Python port (paper/figures/imagegen.py) calling the configured Azure OpenAI image deployment (gpt-image-2.5 family, 1536x1024, quality high) through environment variables ... Response: 3 icon sheets generated (16 icons), each sheet inspected visually: flat key background, no text or arrows, consistent style, QA: 16 pass, 0 warn, 0 ...
E034	figure_rebuild	Figures rebuilt in draw.io (editable): generated icons embedded as raster elements, all text, panels, arrows and chips native. Fig 1 (framework and pipeline) and Fig 2 (interaction modes) exported at 4x PNG and reviewed twice, defects fixed: overlapping text in panel B, icon ... Response: Final figures in paper/figures, flat v0 versions retained as fig*_v0 records
E035	layout_defect	Main text rendered as 7 pages with a 40% blank area on page 2 because Figure 1 and its caption could not fit and the caption style chained keep-with-next to following headings, human asked for exactly 8 pages (I025). Fix: caption keep-with-next off, subtype and boundary-rule ... Response: Main text now fills 8 pages, references on pages 9-10, appendix 8 pages
E036	parallel_call_race	Several tool calls issued in the same block (read of an output file together with the command that writes it) returned stale or missing content, causing a wrongly stale style report to be read once Response: Dependent calls now issued sequentially
E037	style_check	Human-style checklist script run on the manuscript: removed dashes and semicolons (APA multi-citations rewritten as 'following A and B'), split sentences over 32 words and paragraphs over 10 sentences, replaced defensive or negative formulations by positive ones where outside ... Response: Report clean except legitimate uses of 'only' and 'not' in definitions and limitations
E038	open_item	Open items for human: (1) qualitative interpretations in Findings were drafted by the AI and not all cited units were reviewed by the human, (2) end pages of Mintzberg et al., Eisenhardt and Hevner et al. need confirmation, (3) all references must be re-checked with other tools ... Response: Reported to human in final message
E039	method_misreport	AI told the human figures were built with the Sivia method. In fact only generate-figure-elements (sheet prompts, cut-out, split, QA) was run. Skipped: reading the required references (visual grammar, overview narrative, prompt templates, ImageGen-first workflow), Figure Claim ... Response: Disclose, redo with the full workflow if human agrees
E040	prompt_length_gate	Figure 2 mechanism prompt first failed the Sivia length gate (15,227 vs 17,973 non-whitespace characters) before any image call, expanded with concrete lane geometry, card interiors and routing instead of padding. Figure 1 draft v1 (18,230 characters) was generated and ... Response: Prompts revised and rechecked before generation
E041	prompt_edit_failure	An edit meant to remove the title section of the Figure 2 prompt silently did not apply (string mismatch), the AI generated fig2_draft_v1 with a leftover instruction to print a title, noticed it by grepping the prompt after generation, fixed the prompt, rechecked the length gate ... Response: Superseded by v2 drafts
E042	visual_draft	Sivia ImageGen-first stage: whole-figure drafts generated from full instantiated prompts bound to the Sivia overview and mechanism templates (Fig 1: 18,287 vs 15,356, Fig 2: 22,010 vs 17,973 non-whitespace characters). Role reference inspected: knowledge-base case ReAct ... Response: Awaiting human feedback

ID	Type	Event and response
E043	asset_production	Sivia element pipeline run for the approved drafts: 5 key-colour sheets (18 + 12 icons) generated with the Azure image deployment, local key-colour cut-out, split and QA. All 30 assets pass the alpha/clipping/spill/dpi checks (338-567 effective dpi), contact sheets inspected by … Response: Assets accepted, human has not yet reviewed them individually
E044	tool_defect	sivia_assets.py cutout crashed with 'assignment destination is read-only' when --erode > 0 (defect in the external tool, not in project code). Workaround: --erode 0, which produced clean edges per QA. The external tool was not modified Response: --erode 0
E045	method_deviation	The drawio_live_* MCP tools required by the Sivia draw.io Drawer skill are not available in this environment. The editable reconstruction is therefore produced by a Python generator that writes native draw.io cells (text, shapes, edges) plus the 30 atomic icon images, then … Response: Disclose deviation
E046	reconstruction	Editable reconstruction of Figures 1 and 2 as native draw.io files (paper/figures/drawio/fig1.drawio, fig2.drawio) built by dxlib.py and build_fig1/2.py with 30 icon assets. Fixed the four defects the AI had listed (text over gear, unlabelled interaction-mode icons, ambiguous … Response: Awaiting human visual review of final figures
E047	tool_availability	New session with the Sivia draw.io live MCP server attached (drawio_live_ *tools present, unlike the previous session in which they were missing). Image generation credentials (the image service variables* / the image service key variable) are not present in this session, so no … Response: Reuse of existing generated assets, recorded as a deviation from a full regeneration cycle
E048	tool_availability	Correction to E047: the Azure OpenAI image credentials are configured, but they are stored in project .env files outside this session environment, so the earlier statement that no credentials are present was wrong. Live probes on 2026-10-22 show that the [resource] resource with … Response: Keep the reuse decision recorded in E047 because elements_b and elements_c already cover every glyph of the two figures and share one style, call the …
E049	tool_availability	The human recalled a 2.5 image model under the [resource] resource (I033). Live probes on 2026-10-22 show that the model catalogue of both [resource] and [resource] lists gpt-image-2.5-flare and gpt-image-2.5-sunburst, but no deployment exists under those names, so every image … Response: Image generation stays available through gpt-image-2. gpt-image-2.5 is a catalogue entry without a running deployment, so it cannot be used. The …
E050	tool_behaviour	Multi-line labels on the live draw.io path were tested before the figure build: a label written with a newline and a label written with an HTML break both render as a line break. The live canvas was exported to SVG and the generated foreignObject markup was read to confirm this, … Response: Labels can be written with plain newlines in dxlib.mjs
E051	figure_defect	Figure 1 v1 was exported and delivered, and the human review found that the draw.io callout shape (shape=callout with a wavy speech-bubble outline) carried its pointer at the bottom left, that the human and the agent messages were stacked in one column, and that the Tags panel … Response: v2 rebuild of Figure 1 with a plain rounded bubble plus a side tail, a two-sided timeline in panel A, a two-sided panel B and a relaid out Tags …
E052	layout_revision	Figure 1 v2 was rebuilt after the human review. dxlib.mjs now draws a message bubble as a rounded rectangle with a triangle tail on the side of the speaker, the tail being drawn under the body so the body hides its base. Panel A became a chat timeline with the human turns on the … Response: Figure 1 v2 is ready for the human visual check before the paper build is refreshed
E053	figure_build	Figure 2 was built live in the visible draw.io canvas through paper/figures/live/build_fig2.mjs, the first live build of this figure. Panel G lists the six interaction modes as numbered lanes with one type tile per side, a direction arrow that points from the earlier unit to the … Response: live audit, overlap check and empty-area check on paper/figures/drawio/fig2_sivia.drawio
E054	paper_rebuild	The manuscript and the workflow appendix were rebuilt with the new figures. The Figure 2 caption now says that the bubble texts are abridged English renderings of messages in the project, the same wording as the Figure 1 caption, because the recorded messages are in Chinese and … Response: the main paper ends on page 8 and the references start on page 9, so the main paper is within the eight page limit

ID	Type	Event and response
E055	model_switch	The figure rebuild of this session up to the Figure 2 delivery was logged under opencode/big-pickle (MiniMax-Flash) (events E045 to E054). From this turn the work continues with github-copilot/claude-sonnet-5.5 after the human rated the previous figure aesthetics as poor (I036). … Response: recorded here and in records/environment.md
E056	figure_defect	Review of the exported PNGs of Figure 1 v2 and Figure 2 found defects that the earlier gates did not catch. (1) No arrowhead is drawn on any arrow: drawio_live_add_line forces end_arrow to none unless the end_arrow argument is passed, so the endArrow=block written into the style … Response: redesign of both figures with a new design system, end_arrow passed explicitly, the geometry check extended to text cells, every export is viewed …
E057	data_integrity_incident	The sync script overwrote the raw snapshots of 29 sessions (origin ideation session and 28 coder sessions) with the 2,305-byte help text of the opencode CLI, because the CLI had been upgraded and `opencode export <id>` no longer exports. A first repair with `opencode session … Response: All snapshots were rebuilt from the OpenCode database, opened read-only, with records/scripts/export_session_legacy.py. A legacy file (info and …
E058	verification_gap	Residual risk of E057: the pre-incident legacy files were overwritten, so the rebuilt files are not byte-identical to them. Equality of content was shown for one session (figure redraw session, 323 of 323 messages identical in id, order and content, 23 newer messages added) and … Response: Both formats are kept and listed in the registry (v2_file, v2_messages). To be stated in the workflow appendix if the human agrees. The coding corpus …
E059	reporting_audit	Audit of the figure redraw session (ses_e9dd5da0, 2026-10-22 13:51 to 17:28) against its reports. Matches: all six human messages with content are logged (I031 to I036), the model change from big-pickle to claude-sonnet-5.5 at 17:04 is logged (E055 and environment.md), raw … Response: Gaps 1 to 3 closed by this session (E060, A032 to A038, V023 to V028). Gaps 4 to 7 are recorded here and not edited in the append-only logs. A032 …
E060	figure_build	Figure 2 version 2 of the other session was adopted unchanged after review (live audit zero hard failures and warnings, 201 objects, 22 pictures, counts checked against research/coding/final/transitions.csv: modes 1 to 6 give 8/6, 7/6, 1/3, 8/9, 2/4, 39/22). Figure 1 version 4 … Response: Paper rebuilt: body ends on page 8, references on pages 9 and 10, style checker clean apart from names and one long paragraph that existed before. …
E061	correction	E057, E059 and INCIDENT.md did not say who started the two writes that damaged the raw snapshots. Both runs of records/scripts/sync_sessions.ps1 (20:37 and 20:40 on 2026-10-22) were started by the primary agent in this session, the second after a repair that used the new export … Response: INCIDENT.md has a new section on cause and responsibility, and the appendix account names the primary agent. Disclosure in the appendix was approved …
E062	consistency_fix	Figure 2 panel B showed the human unit S01-U032 as Decision while the manuscript describes it as a human Proposal (Findings, RQ1, and the mixed contribution of the typology). The coders disagree on its primary tag (Coder A D, Coder B K) and both tag P, D, K and M. The chip was … Response: Figure 2 rebuilt through the live tools and audited with zero hard failures and warnings, version 2 kept in paper/figures/live/archive and …
E063	source_access	AIS eLibrary, the publisher of ICIS papers, returned HTTP 403 to scripted requests and to the desktop browser (page title: this request was blocked). Web search and the ICIS 2026 site were also unavailable. The block was not circumvented. The benchmark sample was taken from … Response: Sample limits stated in research/structure_review_ICIS.md: convenience sample, author versions, submission type inferred from page count, …
E064	template_rule	The human saw no abstract in the paper. The ICIS 2026 template states that authors and abstracts should not be in the file that contains the body of the paper because they are added in the review system. All 16 sampled ICIS papers have an abstract (median 159 words) because they … Response: paper/submission_form_abstract.txt. If the human wants the abstract inside the PDF this departs from the template
E065	integrity_defect	Walker and Whittaker (1990) was cited in the manuscript text and was missing from the reference list, in the manuscript versions of earlier sessions as well. A cited work without a reference entry is an integrity flag for the automated screening of the call Response: Metadata verified in Crossref (DOI 10.3115/981823.981833) and the entry added. The audit now shows 31 references, all cited, no citation without entry

ID	Type	Event and response
E066	layout_gap	White space at the foot of pages. The Word export showed gaps of 10.7, 6.8 and 8.0 lines on pages 2, 5 and 6 and 4.3 free lines on page 8, because a figure or table that cannot split moves whole to the next page. The human named pages 3, 5 and 6, which suggests a different … Response: Body of 8 pages, references on pages 9 and 10, largest foot gap 4.6 lines, about 2.4 free lines on page 8
E067	style_defect	The AI introduced style defects while merging sentences to save space, five sentences of 33 to 36 words, and used semicolons in grouped APA citations, which the human-style rules forbid Response: Sentences split again and grouped citations written as narrative citations. The check is clean apart from author-name fragments and one pre-existing …
E068	framing_revision	Manuscript restructured as a conceptual paper positioned in IS theory (option A of I042). The title and running header changed, the research questions are set off in a shaded block in plain wording, a Theoretical Background rests on nine verified IS sources, an account of … Response: Previous script archived as paper/archive/build_paper_v2_before_framing_revision.py and build_paper.py now holds the new text. Body fits 8 pages and …
E069	method_deviation	Coding was not repeated for this revision, although the human had said that each larger manuscript update should be recoded and the differences reported (S01-U045). The revision changes the framing and the explanation and leaves units, tags and counts untouched. Units produced … Response: The Case and Data paragraph states that the coded corpus is fixed at 146 units, and this row records the reason. The human decides whether to extend …
E070	provenance_risk	The raw outputs of both coders for the v1.1 runs (16 jsonl files) existed only in a temporary directory of the machine, and analysis_final.py reads them from there Response: Copied the 16 files into research/coding/final/coder_results/ before further use. The original analysis scripts still point to the temporary …
E071	interpretation_risk	The three propositions and the mechanisms of the RQ3 section are the primary agent's interpretation of 21 delegation episodes and 15 AI Proposals read against the codes. The AI defined an episode as a human message plus the AI responses up to the next human message, and read the … Response: Stated in Limitations that the primary agent drafted the interpretations including the mechanisms, and Table 3 gives evidence that would count …
E072	error_caught	In the first draft of the account the evidence cell of proposition P1 said that the two longest runs followed scoped instructions, while the Discussion said that the one overreach arose from an ambiguous instruction. The same instruction (S01-U045) was described both ways. A … Response: P1 was restated as a run that ends with a report on the delegated scope or with a human interruption after the AI acts beyond it, and its evidence …
E073	file_overwrite	While changing one label of Figure 1, the AI ran the first-generation builder paper/figures/drawio/build_fig1.py and finalize.py. They overwrote paper/figures/fig1_final.png and fig2_final.png with the rejected first-generation design and replaced drawio/fig1.drawio and … Response: fig1_final_v4.png and fig2_final_v3.png were copied back to fig1_final.png and fig2_final.png. The banner text of Figure 1 was changed in a copy of …
E074	log_correction	The timestamps of rows E068 to E073, V045 to V057 and A045 to A049 were typed by hand and are wrong. Rows written in the first append carry 12:40 and the later append carries 13:05, while the machine clock read 12:34 at the first append and 12:40 at this entry Response: The logs are append-only, so the rows stay. The real write times are 12:36 to 12:40, and the order of the IDs is the true order. Later rows take the …
E075	unrequested_change	The AI changed the title to 'Who Thinks What in AI-Led Research?', the running header and the central question to 'AI-led research', moved three tables from the main text to the appendix and removed 'Qualitative Case Study' from the title, although the human had asked only for … Response: Title, header and the question wording are restored to the human-AI wording. Tables moved to the appendix (coded excerpt, exemplars, agreement) are …
E076	recode_run2	Recode of the extended corpus under codebook v1.1 (hash 488B8CFC…9C53 for the coder copy), as required by the human rule for larger revisions and requested by the human (I043). Corpus built from the synced exports: 294 units, of which 265 were kept. Exclusions by rule: empty … Response: Seven units removed after coding by rule R3b. Their coder outputs stay in research/coding/final_v2/coder_results and are not used by any table. The …

ID	Type	Event and response
E077	tool_conflict	Builds of Figure 3 failed repeatedly at the reopen and audit step with 'Port 9333 is already in use by a non-draw.io process'. The OpenWork sidecar restarts the registered draw.io live MCP server every few seconds, and each copy claims the port. Background runs of the build were … Response: Builds were run in the foreground after stopping draw.io and the stray live servers, and the saved file was audited on port 9351 with a new script …
E078	page_overflow	After the recode and the new account the body ran to 9 pages and Figures 1 and 2 left blank bands of 8 to 21 lines where they did not fit. The AI shortened the text by about 600 words, moved Figure 1 to the start of Research Design, narrowed Figure 2 and Table 3 and compacted … Response: Body fills pages 1 to 8 with a largest gap of 2.0 lines on page 8, references on pages 9 and 10. Some cut sentences were restored once the section …
E079	unrequested_change	The AI first kept a pointer sentence about generative AI in the Conclusion after the human had said the section may go, because the template carries a generic line on the use of AI. The human confirmed that the form has the item Response: Pointer sentence removed, text kept for the form
E080	error_caught	After the recode, panel A of Figure 2 still showed the transition counts of run 1 (8/6, 7/6, 1/3, 8/9, 2/4, 39/22) while the text and Table 3 used run 2 (9/9, 9/12, 3/4, 17/17, 3/4, 85/62). The mismatch was found by the AI when it compared the figure source with the new … Response: Figure 2 was rebuilt live with the run 2 counts and a smaller bar scale (0.9 instead of 1.45) so that the bar of 85 fits, audited with 0 hard …
E081	record_gap	The 28 coder sessions of run 2 were not captured by the session sync after the recode. The sync takes sessions from the project directory and from records/extra_sessions.txt, and the coders run in a neutral temp directory, so each coder session must be listed by hand. The … Response: The 28 session ids were appended to records/extra_sessions.txt, the sync was run again (69 sessions) and the role of each new coder session was …
E082	privacy_defect	Before the PDF was assembled, a scan of the workflow appendix found four defects in the version built on 26 October. (1) Table A3 held seven rows with Chinese or garbled text (I022, I031 to I036). (2) Cloud resource names that contain the author's first name appeared in the … Response: The appendix generator now masks resource names and variable names and writes the coder prompt in reported speech. The Chinese originals were …
E083	record_defect	The log rows I022, I033 and I035 were written to interventions.csv in a garbled encoding at the time of logging, so their quoted Chinese could not be read. The original wording exists only in the raw sessions Response: Every human message was exported verbatim and in UTF-8 from the raw sessions to records/verbatim/human_messages_verbatim.csv with a hashed snapshot …
E084	disclosure_change	The appendix said that the raw session exports would be released in redacted form, as planned in I008. A scan of the 69 raw sessions (about 130 MB) found about 13,700 occurrences of the user name, 11,900 local paths, 156 email addresses including a third party's private address, … Response: The bundle releases the structured layer instead (units with full text, verbatim human messages, logs, codebooks, scripts, coder outputs). The raw …
E085	figure_tool_note	The page numbers of the appendix started again at 1 in the combined PDF Response: The appendix section now starts at page 11, so the numbers run on after the ten pages of paper and references
E086	release_published	The artifact bundle was pushed to the human's GitHub repository [repository URL] as one commit (main, 275 files). The repository was empty before. The commit author is the neutral identity 'Authors' with a noreply address, line-ending conversion was switched off for the commit … Response: A fresh clone of the repository was compared with MANIFEST.sha256, with 0 mismatches. The human will paste an anonymous mirror link into the artifact …

Table A4. Events, errors and responses.

The Record Incident of 22 October

One error damaged the record itself and is told in full here. The recording script exported each session with the command opencode export and wrote the output to the raw archive without checking it. The command line tool was upgraded after the previous run of the script and before the audit, and the same command then printed its help text. The primary agent ran the script at the start of its audit of the figure redraw

session on 22 October, and the help text replaced the raw files of 29 sessions (E057). The agent saw that every session showed zero messages and an identical file size, and it stopped to investigate.

The first repair used the new export command. It restored the 29 files in a format that omits messages removed by context compaction, and the agent ran the script again before comparing the restored files with the earlier snapshots. That second run wrote the same format over 11 further files, among them the main session. The comparison then showed that the figure redraw session had kept 1.3 thousand characters of assistant text where the earlier export held 68.9 thousand.

The OpenCode database still held every message. The agent opened it read-only and rebuilt every session in the earlier format and in the new format, which holds all messages written since the upgrade. It rewrote the script so that it reads the database, writes both formats and refuses to replace a file with a version that has fewer messages. For the figure redraw session the rebuilt file matched the surviving snapshot in all 323 messages. For the other 40 sessions only the message counts could be compared with the counts logged before the incident (E058). The superseded files are kept in the repository, and the coding corpus of 146 units had been built before the incident.

The incident also bears on the argument of the paper. The error arose in work that the AI performed, the AI found it while carrying out a human request to check another session (I037), and the AI made the first repair mistake. The human then decided that the incident should be disclosed in this appendix (I039). The case shows detection and repair by the agent, and it shows the human holding the decision about what the record says.

Time on 22 Oct	Step	Result
17:11	Last successful run of the script before the upgrade	All sessions exported in the earlier format.
20:37	Run of the script by the primary agent at the start of the audit	29 raw files replaced by help text, zero messages (E057).
20:39	First repair with the new export command	29 files restored in a format without compacted messages.
20:40	Second run of the script	11 further files, among them the main session, written in that format.
20:49	Comparison of the two formats and rebuild from the database	Earlier format rebuilt for 41 sessions, identity checked on one session (V023, V024).
20:51	Script rewritten and run again	Both formats written, no file replaced by a shorter one.

Table A5. Timeline of the record incident.

F. Verification and Validation

ID	Target	Method	Result and detail
V001	sources_candidates.csv S001-S075	bibliographic_match	verified_bibliographic_only. 75 candidate sources: title/authors/year/venue matched a real record, content notes come from model memory EXCEPT rows flagged abstract_read. NOT eligible for manuscript claims about content until read
V002	S035 Vaccaro et al. 2024, S045 Gottweis et al. 2026	independent_recheck	verified. Parent agent re-fetched both DOIs from Crossref, metadata match sub-agent claims, abstracts read. S035: 106 experiments/370 effect sizes, human-AI combos worse than best single (g=-0.23), losses in decision tasks, gains in content creation. S045: …
V003	research/fulltext/S029.md (Garrison 2001)	full_text_read	read. Parent read methods/results: message unit, indicators/descriptors, code up/down, kappa .35/.49/.74, CR .45/.65/.84, 95 messages
V004	research/fulltext/S008.md S049.md S068.md S071.md	spot_check_of_subagent_notes	plausible_specific. Notes contain specific numbers, locators and stated limits and flag unread parts, not re-verified against the sources themselves

ID	Target	Method	Result and detail
V005	research/source_access_status.csv	access_summary	done. FULLTEXT_READ 25, PARTIAL 9, ABSTRACT_ONLY 12, NOT_ACCESSED 8 (before human-supplied PDFs)
V006	unit definition (response) on raw exports	dry_run	done. 499 assistant messages across 12 sessions: 72 main (35 text, 37 tool-only), 427 sub-agent (40 text, 387 tool-only), tool-only steps must attach to a text-bearing unit
V007	research/codebook_v1.0_FROZEN.md	freeze_hash	recorded. d9fc917f6a6c9b520b92f51d8250465d836d5af4f883a07cb340e6e3785e7d9a
V008	research/codebook_v1.1_FROZEN.md	freeze_hash	recorded. 2ccb4592396328910b92fc43ff6b6ae6a2df410a45f36c4dedadedab8a8fc266
V009	paper/ICIS_PDW_paper.docx	style_checklist	clean. No dashes, semicolons or multi-colon paragraphs, remaining long-sentence hits are citation splits
V010	paper/ICIS_PDW_paper.pdf	page_count	done. Main text pages 1-8, references pages 9-10
V011	paper/references.txt	bibliographic_match	done. 23 references, authors taken from records, end pages of 3 entries need human confirmation
V012	research/coding/final/agreement_v1.1_all146.csv	agreement_statistics	done. Percent agreement, kappa and alpha with bootstrap CI for 13 actor-type variables over 146 units
V013	Azure image endpoints	image_generation_probe	done. gpt-image-2 on [resource] with api-version 2024-02-01 returns an image in 19.8 s (337 KB), gpt-image-2.5-flare and gpt-image-2.5-sunburst exist only in the catalogue and answer DeploymentNotFound
V014	paper/figures/live/probe_multiline.mjs	live_canvas_probe	done. A label written with a newline and a label written with an HTML break both render as a line break in the exported SVG foreignObject markup
V015	paper/figures/drawio/fig1_sivia.drawio	live_canvas_audit	done. Figure 1 v2: page 1440 x 1000 visible, 275 objects (250 vertices, 25 edges, 54 pictures), zero hard failures and zero warnings
V016	paper/figures/live/check_geometry.mjs	geometry_check	done. No two payloads overlap in Figure 1 v2, the only intentional contacts are a panel or card holding its content and a bubble tail meeting its body
V017	paper/figures/live/check_blanks.mjs	blank_area_check	done. Content row share per panel in Figure 1 v2: A 87 percent with a largest empty band of 12 px, B 87 percent, C 99 percent, D 94 percent, E 87 percent, F 85 percent with a largest empty band of 21 px
V018	paper/figures/drawio/fig2_sivia.drawio	live_canvas_audit	done. Figure 2: page 1440 x 900 visible, 143 objects (126 vertices, 17 edges, 10 pictures), zero hard failures and zero warnings
V019	paper/figures/live/check_geometry.mjs	geometry_check	done. Figure 2 has no overlapping payloads, the first run reported four collisions between the card number badge and the bubble tail in panel H, which were removed by narrowing the card bubbles and moving the badges
V020	paper/figures/live/check_blanks.mjs	blank_area_check	done. Figure 2 content row share per panel: G 69 percent with a largest empty band of 28 px, H 81 percent with 28 px, I 82 percent with 30 px
V021	paper/ICIS_PDW_paper.pdf	pdf_check	done. The rebuilt manuscript has 10 pages, the main text ends on page 8 and the references occupy pages 9 and 10, so the main paper meets the eight page limit of the call for papers
V022	paper/figures/live/check_blanks.mjs	blank_area_check	done. Figure 1 v2 content row share per panel: A 87 percent with a largest empty band of 12 px, B 87 percent, C 99 percent, D 94 percent, E 87 percent, F 85 percent with 21 px
V023	records/raw_sessions/ (41 files)	snapshot_restoration	done. All 41 sessions rebuilt in legacy format and dumped in v2 format. Message counts of the 40 earlier sessions equal those in snapshot_log.csv before the incident

ID	Target	Method	Result and detail
V024	ses_e9dd5da0 legacy rebuild	content_comparison	done. 323 of 323 messages identical in id and order, one message differs because it was still being written, 23 newer messages added, 1 part differs only in tool state
V025	paper/figures/drawio/fig1_sivia.drawio	live_canvas_audit	done. Figure 1 version 4: page 1280 x 960, 288 objects (261 vertices, 27 edges, 31 pictures), zero hard failures and zero warnings after two correction rounds from 7 hard failures
V026	paper/figures/drawio/fig2_sivia.drawio	live_canvas_audit	done. Figure 2 version 2: page 1280 x 744, 201 objects (171 vertices, 30 edges, 22 pictures), zero hard failures and zero warnings. Counts equal transitions.csv
V027	Figure 1 version 4	geometry_and_blank_check	done. Only the nine intended bubble-tail overlaps of 3 px. Content share per panel 79 to 94 percent, largest empty band 21 px. Grey-scale render shows every type still readable by its letter. Print-size render (6.15 in) legible
V028	paper/ICIS_PDW_paper.pdf	pdf_and_style_check	done. Body ends on page 8 (Conclusion and Use of Generative AI), references on pages 9 and 10, Figure 1 on page 3 and Figure 2 on page 7. No dash, semicolon or colon violation. One 11-sentence paragraph and four author-name fragments flagged as before
V029	paper/figures/drawio/fig2_sivia.drawio	live_canvas_audit	done. Figure 2 version 3: 201 objects, zero hard failures and zero warnings. Export viewed
V030	paper and appendix PDFs	pdf_and_style_check	done. Paper body ends on page 8, references on pages 9 and 10. Appendix has 13 pages. Style check clean for the paper apart from names and one pre-existing long paragraph. In the appendix the only findings are pre-existing long sentences in Sections C and G. …
V031	Benchmark sample of ICIS papers	comment_field_check	done. 16 arXiv papers whose comment names the AIS International Conference on Information Systems (ICIS 2019 to 2026). Papers of ICISSP, IEEE ICIS, ISD and Pre-ICIS workshops were excluded. Three comments carry an AISeL link
V032	Lu et al. (2024) The AI Scientist	bibliographic_match	done. Title and six authors match. Claims used (end-to-end pipeline, human share unmeasured) come from the reading note S043, which is partial
V033	Risko and Gilbert (2016) Cognitive offloading	bibliographic_match	done. Metadata match. Claim used (offloading follows a metacognitive evaluation) comes from the full-text note S017, read from the author manuscript
V034	Gilardi et al. (2023) ChatGPT outperforms crowd workers	bibliographic_match	done. Metadata and article number e2305016120 match. Claim used (codebook as instruction for the LLM annotator) comes from note S065, read from the preprint
V035	Xiao et al. (2023) Supporting qualitative analysis with LLMs	bibliographic_match	done. Metadata, IUI 2023 and pages 75-78 match. Claims used (codebook-based prompt, lower agreement with experts than experts among themselves) from note S068
V036	Pangakis et al. (2023) Automated annotation requires validation	bibliographic_match	done. Title and authors match. Claim used (validate against human labels) from note S067
V037	Ziems et al. (2024) Can LLMs transform computational social science	bibliographic_match	done. Metadata match, Computational Linguistics 50(1) 237-291. Claim used (LLMs not a replacement for human annotators) from note S066
V038	Walker and Whittaker (1990) Mixed initiative in dialogue	bibliographic_match	done. Metadata and pages 70-78 match. The Crossref title omits the subtitle, which the ACL Anthology record in note S058 gives
V039	ICIS 2026 PDW call for papers	local_copy_check	done. Organiser name, title and site taken from the PDF metadata and first page. The nine functions of the contribution account are quoted from the submission guidelines

ID	Target	Method	Result and detail
V040	Citation audit of the manuscript	script_check	done. 31 references, every one cited in the body. After adding Walker and Whittaker no citation without a reference entry remains. Two flags are artefacts of the script (a particle in de Jong and an organisation author)
V041	Manuscript layout and style	pdf_and_style_check	done. Body ends on page 8, references on pages 9 and 10, largest foot gap 4.6 lines. Style check clean for dashes, semicolons and colons. The abstract has 160 words and passes the same checks
V042	Gregor (2006) The nature of theory in information systems. MIS Quarterly 30(3), 611-642	crossref_lookup	verified. DOI 10.2307/25148742 resolves to the stated article. The abstract confirms five theory types including theory for analyzing. That taxonomies belong to the first type is from the AI's memory and was not checked in the full text. Doty and Glick (1994) ...
V043	Klein and Myers (1999) A set of principles for conducting and evaluating interpretive ...	crossref_lookup	verified. DOI 10.2307/249410 resolves to the stated article and authors
V044	Webster and Watson (2002) Analyzing the past to prepare for the future: writing a ...	crossref_lookup	verified. Title, authors, journal, volume, issue and pages match
V045	Gregor (2006) The nature of theory in information systems. MIS Quarterly 30(3), 611-642	full_text_read	read. Read the abstract and the sections on Type I Theory for Analyzing (classification schema, frameworks and taxonomies are variants) and Type II Theory for Explaining (explains how and why, may be process-type theory). The copy carries the MIS Quarterly ...
V046	Baird and Maruping (2021) The next generation of research on IS use. MIS Quarterly 45(1), ...	crossref_lookup	verified. Title, authors, volume, issue and pages match. The abstract supports the statements that agentic artifacts can assume responsibility for tasks with ambiguous requirements and that delegation is proposed as the lens
V047	Fugener, Grahl, Gupta and Ketter (2022) Cognitive challenges in human-artificial ...	crossref_lookup	verified. Metadata match. The abstract supports the statement that humans judge their own metaknowledge poorly and delegate knowledge work to AI poorly. The article appeared online in 2021 and in the 2022 issue
V048	Berente, Gu, Recker and Santhanam (2021) Managing artificial intelligence. MIS Quarterly ...	crossref_lookup	verified. Metadata match. The abstract names autonomy, learning and inscrutability as the three facets on which decisions are made
V049	Seeber et al. (2020) Machines as teammates: A research agenda on AI in team ...	crossref_lookup	verified. Metadata match and the twelve authors listed in Crossref are all in the reference. The abstract confirms a research agenda for machines as teammates
V050	Gronsund and Aanestad (2020) Augmenting the algorithm. Journal of Strategic Information ...	crossref_lookup	verified. Metadata match. The abstract supports the statements that humans and the algorithm were configured and reconfigured and that humans audited and altered the algorithm
V051	Leonardi (2011) When flexible routines meet flexible technologies. MIS Quarterly 35(1), ...	crossref_lookup	verified. Metadata match. The abstract supports the use of imbrication of human and material agencies. The manuscript uses the idea only as an analogy
V052	Susarla, Gopal, Thatcher and Sarker (2023) The Janus effect of generative AI. Information ...	crossref_lookup	verified_bibliographic_only. Metadata match. The Crossref and OpenAlex records hold no abstract, only a funding note, so the claim in the manuscript rests on the title
V053	Benbasat, Goldstein and Mead (1987) The case research strategy in studies of information ...	crossref_lookup	verified. Metadata match. The abstract confirms that the article defines the case research strategy for IS
V054	Doty and Glick (1994) Typologies as a unique form of theory building. Academy of ...	crossref_lookup	verified. DOI 10.5465/amr.1994.9410210748 (also listed as 10.2307/258704) resolves to the stated article. Crossref gives the first page only. Gregor (2006) cites it for criteria on classification schema

ID	Target	Method	Result and detail
V055	Numbers and quotations of the RQ3 section	script_assertion_and_unit_reading	verified. The counts (21 episodes, median run of 2, four long runs holding 43 of 75 responses, 40 and 39 of those 43 coded Execution or Metacognition, 13 of 15 human Decisions after AI Proposals) are asserted in the build and fail it if they differ. The unit ...
V056	Manuscript layout, style and citations after the framing revision	pdf_and_style_check	clean. The body ends on page 8 with 1.0 line to spare and references fill pages 9 and 10. The style check reports no dashes, semicolons or multiple colons, and its remaining items are splits at et al. and the risky-phrase list. The citation audit finds 40 ...
V057	Figure 1 v5 against v4	pixel_comparison	verified. After cropping the export to the framing of v4 the images differ in 15,014 pixels, all inside the bounding box of the bottom banner (x 733 to 1800, y 1817 to 1854). The new text fits inside its frame
V058	Recode run 2: 265 units, both coders	completeness_and_integrity_check	verified. Both coders returned all 272 units of the first corpus and 0 bad lines. 265 units after post-coding removal. Run 2 outputs archived before use. Agreement recomputed with the earlier definitions (percent, kappa, alpha, 2,000 bootstrap draws)
V059	Numbers stated in the manuscript	script_assertion	verified. Unit counts, existence table, transitions per mode, 38 episodes, 11 long runs holding 120 of 177 responses, 113 and 110 Execution or Metacognition, 19 and 18 of 21 Decisions after AI Proposals, 90% and 95% repeat of primary type, 129 units after the ...
V060	Endings of the 9 long runs followed by a human message	unit_text_reading	verified. Five runs end with a report of more than 1,000 characters (S01-U013, U031, U086, U133, U149) and four end in short working steps that the human then stopped, rejected, corrected or extended (S01-U063, U106, U125, U163, answered by S01-U064, U107, ...
V061	Figure 3 audit	live_canvas_audit	clean. 70 objects, 0 hard failures, 0 warnings after one hard finding (a tick crossing the document icon) was repaired. Quotes of the figure are abridged English renderings of S01-U020, S01-U021 to S01-U031, S01-U031 and S01-U032
V062	CFP requirement of a statement on the use of generative AI in the paper	document_check	verified. No CFP text requires the statement in the paper. The submission site provides a Human-AI Contribution Disclosure Form and an Artifact Availability Statement, and the workflow appendix must hold a human-AI contribution account. The template has a ...
V063	Manuscript after the recode and the layout work	pdf_and_style_check	clean. Body fills pages 1 to 8, largest gap 2.0 lines on page 8, references on pages 9 and 10. No dashes, semicolons or multiple colons. No sentence above 32 words. 40 references, all cited
V064	Figure 2 rebuilt with run 2 counts	live_canvas_audit	clean. 201 objects, 0 hard failures, 0 warnings. The six pairs of counts equal the sums of the transitions in research/coding/final_v2/transitions.csv: 9/9, 9/12, 3/4, 10+7=17 and 12+5=17, 3/4, 74+11=85 and 53+9=62
V065	Submission PDF (31 pages)	pdf_leak_and_structure_scan	clean. No user name, resource name, organiser, local path, email, key, garbled character, instruction to an AI system, zero-width character or hidden-style text. Metadata holds the title only. Main paper ends on page 8, references start on page 9, appendix on ...
V066	Artifact bundle (274 files, 17.6 MB)	redaction_and_leak_scan	clean. First scan found 46 hits. They were literature surnames, regular expressions inside scripts, a variable named after a resource, and the truncated user name [local path] left by a 90-character cut of tool summaries. The rules were extended and the scan ...
V067	Verbatim archive	hash_snapshot	recorded. 75 user messages (65 human, 10 prompts to sub-agents) from 13 sessions exported in UTF-8, with SHA-256 sums of the four logs, the registry and the export in records/verbatim/logs_snapshot_*

ID	Target	Method	Result and detail
V068	Published repository equals the redacted bundle	clone_and_manifest_check	verified. Fresh clone of the public repository: 275 files, 0 manifest mismatches, commit author 'Authors', email placeholder present in records/verbatim, leak scan of the bundle 0 hits including the new email-body patterns
V009	Manuscript text	Style checklist	Searched for dashes, semicolons, multiple colons, short and long sentences and defensive phrases. Findings were fixed and the check was rerun until clean.
V010	Manuscript layout	Page count	Rendered to PDF with a word processor. Main text fills 8 pages, references start on page 9.
V011	Reference list	Bibliographic match	23 references matched in Crossref, DataCite or OpenLibrary. Author lists were taken from the records. End pages of three entries (Mintzberg et al., Eisenhardt, Hevner et al.) are missing or differ in the Crossref record and need human confirmation. The list now holds 40 references, and the entries added later are verified in their own rows (V042 to V054).
V012	Coding agreement	Statistics	Percent agreement, Cohen kappa and Krippendorff alpha with 2,000-resample bootstrap intervals for each actor-type presence variable. Cross-model coding with two model families.

Table A6. Verification steps. Human re-verification of references with other tools is planned before submission.

G. Data, Code and Artifact Availability

The artifact bundle holds the unit tables with the full text of every coded unit and the redacted verbatim human messages. It also holds the four structured logs, the session registry and the coder outputs of every run. It further holds codebook versions 1.0 and 1.1 with SHA-256 hashes, the analysis and build scripts and the figure sources with generation prompts and quality reports. It holds the structure review against ICIS papers and the project rules and logging skill. Codebook v1.0 has hash d9fc917f6a6c9b520b92f51d8250465d836d5af4f883a07cb340e6e3785e7d9a and v1.1 has hash 2ccb4592396328910b92fc43ff6b6ae6a2df410a45f36c4dedadedab8a8fc266. The stable link is given in the artifact availability statement of the submission form. The raw session exports (69 sessions, about 130 MB) are withheld. Their tool outputs contain local paths, user and resource names and forwarded third-party email text, and a pattern-based redaction cannot make them safe with certainty. They can be provided to reviewers on request. Copyrighted source copies read during the project, among them the repository copy of Gregor (2006), are not redistributed. In the released copies, user names, local paths, email addresses, an organiser's name and cloud resource names are masked, and the Chinese wording of the human messages is kept. The unmasked verbatim record is held privately with SHA-256 sums.

H. Human-AI Contribution Account

Function	Human	AI
Problem formulation	Proposed the topic, restricted the data to the project, formulated the first research questions in conversation and, after a co-author's comment, redirected them toward plain wording and theoretical abstraction.	Assessed fit and risks, offered candidate questions, refined definitions and rewrote the question block.
Literature identification	Required a survey first, demanded original texts and skipped closed sources.	Ran the surveys, verified bibliographic records, read open texts and wrote notes.
Theory development	Proposed the first three-way split, reframed knowledge as a type and renamed expression or design as Proposal.	Synthesised the cognition and decision literature, drafted the typology and boundary rules, searched and verified the IS background and drafted the delegation and commitment propositions.
Research design	Fixed the unit as one model response, the qualitative design, the coder models and the criterion for mixed contributions.	Operationalised the unit, built the coding pipeline and the version-freeze procedure.

Function	Human	AI
Analysis	Ruled on example cases and boundary cases.	Ran the coders, computed agreement and carried out the qualitative analysis.
Interpretation	Adjudicated definitions and examples and judged the three propositions reasonable. The mechanisms have not been reviewed unit by unit.	Drafted all interpretations of existence, distribution, transitions and mechanisms.
Writing	Set structure, style constraints and the page target and rejected several versions. Typed no text of the manuscript or this appendix (I049).	Wrote all text of the manuscript and this appendix and produced every figure and table.
Verification	Halted premature drafting and will re-check references with other tools.	Checked references and sub-agent notes and ran the style and layout checks.
Revision	Rejected several figure versions and the page layout, decided that the record incident be disclosed, relayed the co-author's comment, chose the conceptual restructuring, rejected the box and grid tables, required a recode and full pages, and dropped the AI-use section.	Rebuilt the figures, repaired the record, extended the corpus and revised every version.

Table A7. Contribution account by research function.

I. Coding Agreement and Coded Examples

This section holds the tables that were moved out of the main text. Table A8 gives agreement between the two coders on actor-type presence for all 265 units of run 2. Table A9 shows five consecutive units with the tags of both coders, primary tag first. Table A10 gives one exemplar unit for every actor-type pair that both coders tag. Table A11 compares the two runs on the 146 units that both coded.

Actor and type	GPT / Gemini	Agree %	Kappa [95% CI]	Alpha [95% CI]
Human, Knowledge	12 / 12	100	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
Human, Proposal	3 / 4	100	.85 [.00, 1.00]	.86 [.00, 1.00]
Human, Decision	32 / 29	98	.91 [.81, .98]	.91 [.81, .98]
Human, Metacognition	26 / 28	99	.92 [.82, .98]	.92 [.82, .98]
Primary agent, Knowledge	99 / 42	77	.44 [.33, .55]	.41 [.28, .53]
Primary agent, Proposal	23 / 38	94	.69 [.54, .82]	.69 [.53, .82]
Primary agent, Decision	23 / 8	91	.22 [.03, .42]	.21 [.01, .42]
Primary agent, Metacognition	113 / 111	96	.91 [.85, .95]	.91 [.85, .95]
Primary agent, Execution	68 / 118	81	.60 [.51, .69]	.59 [.48, .69]
Sub-agents, Knowledge	14 / 10	99	.83 [.60, .97]	.83 [.60, .97]
Sub-agents, Proposal	1 / 1	100	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
Sub-agents, Metacognition	17 / 12	98	.82 [.62, .96]	.82 [.62, .96]
Sub-agents, Execution	19 / 32	95	.72 [.56, .85]	.72 [.55, .85]

Table A8. Agreement between the coders on actor-type presence (265 units, run 2).

Unit	Actor	Content (abridged)	Coder A	Coder B
S01-U031	Primary agent	Reports 25 sources read in full and that sub-agent notes were not all re-checked.	K (M, P)	K (M, P)
S01-U032	Human	Argues that agents now give experts new knowledge, so knowledge should count as a contribution.	D (K, M, P)	D (K, M, P)
S01-U033	Primary agent	Announces that it will log the instruction and write the framework to file.	X	X
S01-U034	Primary agent	Delivers framework and codebook v0.1 with knowledge as a type and four primary types.	P (K, M)	P (K, M)
S01-U035	Human	Accepts that evaluation and choice are hard to separate and sets the coding unit to one round.	D (K, M)	D (K, M)

Table A9. Five consecutive units with the tags of both coders, primary tag first (K Knowledge, P Proposal, D Decision, M Metacognition, X Execution).

Actor	Type	Exemplar unit
Human	Knowledge	Execution carries little cognition because it follows a prepared blueprint (S01-U005)
Human	Proposal	A split of cognitive contribution into decision, knowledge and expression or design (S01-U014)
Human	Decision	Only the project's own sessions serve as data (S01-U005)
Human	Metacognition	Survey the related research before designing the scheme (S01-U017)
Primary agent	Knowledge	Reading of the call for papers, including the page limit (S01-U004)
Primary agent	Proposal	A process model linking Knowledge, Proposal, Decision and Metacognition (S01-U066)
Primary agent	Decision	Assignment of a literature survey on forms of cognition to a sub-agent under the project rules (S03-U001)
Primary agent	Metacognition	Report that only abstracts could be read for 12 sources (S01-U031)
Primary agent	Execution	Launching both coders as instructed (S01-U051)
Sub-agents	Knowledge	Reading note that a revised taxonomy is a two-dimensional table (S06-U005)
Sub-agents	Metacognition	Statement that only the main text of one source was read in full (S08-U003)

Table A10. One exemplar unit for each actor-type pair that both coders tag.

Actor and type	Kappa run 1	Kappa run 2	Change
Human, Knowledge	1.00	1.00	0
Human, Proposal	.66	.85	+.20
Human, Decision	.97	.94	-.03
Human, Metacognition	.89	.97	+.07
Primary agent, Knowledge	.76	.80	+.05
Primary agent, Proposal	.84	.97	+.13
Primary agent, Decision	.57	.26	-.31
Primary agent, Metacognition	.91	.86	-.05
Primary agent, Execution	.77	.79	+.02
Sub-agents, Knowledge	.82	.82	0
Sub-agents, Proposal	.66	1.00	+.34
Sub-agents, Metacognition	.61	.81	+.20
Sub-agents, Execution	.66	.69	+.04
Coders agree on primary type (run 1 | run 2)	112 of 146 (77%)	122 of 146 (84%)	
GPT against its own run 1 (primary type | set of types)	90%	79%	
Gemini against its own run 1 (primary type | set of types)	95%	79%	

Table A11. Run 1 and run 2 on the 146 units that both coded: kappa between the coders for each actor-type presence variable, and the repeat rate of each coder against its own run 1.